

Riemannian and Complex Geometry

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— Lecture Notes —

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Preface

We're excited to share these notes as a companion to the course on Riemannian and Complex Geometry. The course offers a hands-on introduction to two fascinating and closely connected areas of differential geometry, both of which provide essential tools for understanding the shape and structure of spaces known as manifolds.

The main goal is to introduce the core ideas and methods used in both Riemannian and complex geometry, while also exploring how they relate to and differ from each other. The course is designed to build a clear and connected understanding of these topics, and to show how they appear in both pure mathematics and theoretical physics.

This edition of the course ran from September 2025 to January 2026, guided by Daniel Galviz and Bowen Liu, as part of a broader effort to support advanced education and collaboration in mathematics across Latin America and beyond.

We want to make it clear that these notes are not meant to be an original work, but rather a well-organized and thoughtful collection of materials and references that support and expand on what was covered in the lectures. Throughout these pages, we've brought together ideas and insights from a range of respected sources that have helped shape our current understanding of geometry.

We are very grateful to the institutions and individuals whose support made this course and these notes possible. Special thanks to the Abdus Salam International Centre for Theoretical Physics (ICTP) – Physics Without Frontiers, for their commitment to building strong academic connections and educational opportunities around the world.

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Bowen Liu wishes to express his thanks to Chenglong Yu and Mao Sheng for their support and guidance, which played an important role in shaping this material and helping bring the course to completion.

Both authors are especially thankful to the Yau Mathematical Sciences Center at Tsinghua University, whose support was essential in making this effort a success. A heartfelt thanks to Shing-Tung Yau, who allowed both of us to enter the beautiful world of mathematics and geometry.

Finally, we thank all the students who joined the course. Their curiosity and dedication greatly enriched the experience. We hope these notes serve as a useful guide for those deepening their understanding of differential geometry, and as a friendly starting point for newcomers to Riemannian and complex geometry. Above all, we hope this material inspires future students and researchers to keep exploring and expanding the frontiers of modern geometry.

Physics Latam, 2025

Daniel Galviz
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Part I

Quick Review on Differential Geometry

1 Smooth manifolds

1.1 Topological spaces

The notion of manifolds is built upon topological spaces. The concept of topological spaces provides open and closed sets that allow defining concepts such as continuity, connectedness, convergence and so on.

Definition 1.1. Let X be a set and $\mathcal{T}$ be a collection of subsets of X satisfying:

- (1) $\emptyset$ and X are in $\mathcal{T}$;
- (2) For any subcollection $\{U_i \in \mathcal{T} \mid i \in I\}$, the union $\bigcup_i U_i \in \mathcal{T}$;
- (3) If $U_1, U_2 \in \mathcal{T}$, then $U_1 \cap U_2 \in \mathcal{T}$.

Then, $\mathcal{T}$ is called a *topology* and $(X, \mathcal{T})$ is called a *topological space*.

Notation 1.2. We usually denote X as a topological space for brevity, if there is no ambiguous.

Definition 1.3. Let $(X, \mathcal{T})$ be a topological space.

1. An element of $\mathcal{T}$ is called an *open set*;
2. A subset V of X is *closed* if its complement in X is an open set, i.e., $X - V \in \mathcal{T}$.

Lemma 1.4.

- (1) For a collection $\{V_i \mid i \in I\}$ of closed sets, the intersection $\bigcap_i V_i$ is also closed.
- (2) For closed sets V_1, V_2 , $V_1 \cup V_2$ is closed.

Definition 1.5. A family of (open) sets $\{U_i\}_{i \in I}$ of X is called an (*open*) *cover* of X if $\bigcup_{i \in I} U_i = X$.

Definition 1.6. A topological space is *compact* if every open cover of X has a finite subcover, i.e., if $X = \bigcup_{i \in I} U_i$ for a collection of open sets $\{U_i \mid i \in I\}$, then we can find finitely many U_{i_k} ($k = 1, \dots, n$) such that $X = \bigcup_{k=1}^n U_{i_k}$.

Definition 1.7. Let $(X, \mathcal{T})$ and $(X', \mathcal{T}')$ be two topological spaces.

- (1) Given a map $f: X \rightarrow X'$, it is called a *continuous map* if $f^{-1}(V) \in \mathcal{T}$ for all $V \in \mathcal{T}'$.
- (2) If f is bijective and both f and f^{-1} are continuous, then f is called a *homeomorphism*.

Definition 1.8. A topological space X is *connected* if it cannot be written as $X = X_1 \cup X_2$, where X_1 and X_2 are non-empty open sets and $X_1 \cap X_2 = \emptyset$. Otherwise, X is called *disconnected*.

Definition 1.9. A topological space is *path-connected* if for every pair of points x and x' in X , there is a path in X from x to x' , i.e., a continuous map $\gamma: [0, 1] \rightarrow X$ such that:

$$\gamma(0) = x \quad \text{and} \quad \gamma(1) = x'.$$

Definition 1.10. Let $(X, \mathcal{T})$ be a topological space and $S \subseteq X$ be a subset. The *subspace topology* on S is defined as

$$\mathcal{T}_S = \{U \cap S \mid U \in \mathcal{T}\}.$$

Example 1.11. Let $\mathbb{S}^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$ with the subspace topology from $\mathbb{R}^2$. Take any pair of points $p, q \in \mathbb{S}^1$. We can construct a function:

$$\gamma(t) = (\cos \theta(t), \sin \theta(t)), \quad \theta(t) = (1-t)\theta_0 + t\theta_1,$$

where θ_0 and θ_1 are the angles corresponding to p and q . This gives a continuous path on $\mathbb{S}^1$, so $\mathbb{S}^1$ is path-connected.

Definition 1.12. Let $(X, \mathcal{T})$ be a topological space and $\sim$ an equivalence relation on X . We define a topology on the quotient set $X/\sim$ by:

$$\mathcal{T}' = \{U \subseteq X/\sim \mid \pi^{-1}(U) \in \mathcal{T}\},$$

where $\pi: X \rightarrow X/\sim$ is the projection map. Then $(X/\sim, \mathcal{T}')$ is the *quotient space* of $(X, \mathcal{T})$.

Example 1.13. Let $X = \mathbb{R}^2$ and $W = \{(x, 0) \mid x \in \mathbb{R}\}$. Then X/W is the set of horizontal lines in the plane. Two points (x_1, y_1) and (x_2, y_2) are equivalent if they have the same y -value, regardless of x . Thus $(x_1, y_1) \sim (x_2, y_2)$ if and only if $y_1 = y_2$. The new space X/W is homeomorphic to $\mathbb{R}$.

Definition 1.14. Let $X_1, \dots, X_k$ be topological spaces. Their Cartesian product is $X_1 \times \dots \times X_k = \{(x_1, \dots, x_k) \mid x_i \in X_i\}$. The *ith projection map* is

$$\pi_i: X_1 \times \dots \times X_k \rightarrow X_i, \quad \pi_i(x_1, \dots, x_k) = x_i.$$

The *product topology* on $X_1 \times \dots \times X_k$ is the topology having as a basis all subsets of the form

$$U_1 \times \dots \times U_k, \quad U_i \subseteq X_i \text{ open}.$$

The space $X_1 \times \dots \times X_k$ with this topology is called a *product space*, and its basis elements are called *product open subsets*.

Definition 1.15. A topological space is called a *Hausdorff space* if for every pair of distinct points x_1, x_2 in X , there exist neighborhoods U_1 and U_2 of x_1 and x_2 respectively that are disjoint.

Definition 1.16. An open cover $\{U_\alpha\}$ of M is *locally finite* if every point $p \in M$ has a neighborhood intersecting only finitely many U_α .

Definition 1.17. A (open) cover $\{V_\beta\}$ is an *(open) refinement* of $\{U_\alpha\}$ if each V_β is contained in some U_α .

Definition 1.18. A topological space is called *paracompact* if every open cover has an open refinement that is locally finite.

Example 1.19. Let X be an infinite set, and τ the topology where open sets are:

- $\emptyset$;
- Any subset $U \subseteq X$ such that $X \setminus U$ is finite.

This topology is called the *finite complement topology*. Take two distinct points $x, y \in X$. Suppose we want to find disjoint open sets $U \ni x, V \ni y$. Since $X \setminus U$ and $X \setminus V$ are finite, their union is finite. Thus:

$$X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$$

is finite. Since $U \cap V \neq \emptyset$, it is impossible to separate x, y with disjoint open sets. Therefore this topology is not Hausdorff.

Remark 1.20. For the case $X = \mathbb{A}^n$ is an affine space, the finite complement topology is exactly the Zariski topology on $\mathbb{A}^n$, which plays an important role in algebraic geometry.

Example 1.21. Take two copies of $\mathbb{R}$, denoted $\mathbb{R}_1$ and $\mathbb{R}_2$. Identify each point $x \in \mathbb{R} \setminus \{0\}$ in $\mathbb{R}_1$ and $\mathbb{R}_2$. Thus $x_1 \sim x_2$ for all $x \neq 0$, but the points $p = 0_1 \in \mathbb{R}_1$ and $q = 0_2 \in \mathbb{R}_2$ are distinct. The points p and q are distinct in the topological space X , but every neighborhood of p intersects every neighborhood of q because they share all points except the origin. Therefore there are no disjoint neighborhoods around these points, so this topology is not Hausdorff.

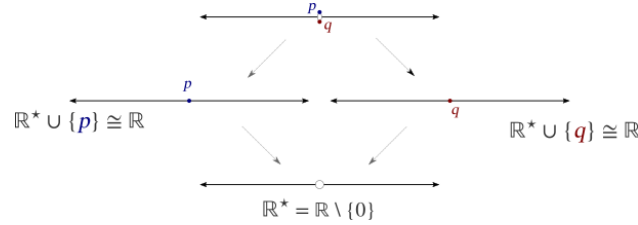


Figure 1: The line with two origins is a non-Hausdorff manifold

Example 1.22. The space $\mathbb{R}^n$ is paracompact. Let $X = \mathbb{R}^n$, and let $\mathcal{A}$ be an open covering of X . Set $\mathbb{B}_0 = \emptyset$, and for each positive integer m , let $\mathbb{B}_m$ denote the open ball of radius m centered at the origin. Given m , choose finitely many elements of $\mathcal{A}$ that cover $\mathbb{B}_m$, and intersect each one with the open set $X - \mathbb{B}_{m-1}$. Let this finite collection of open sets be denoted $\mathcal{C}_m$. Define $\mathcal{C} = \bigcup_{m=1}^{\infty} \mathcal{C}_m$. Then $\mathcal{C}$ is a refinement of $\mathcal{A}$.

It is locally finite: indeed, the open set $\mathbb{B}_k$ intersects only finitely many elements of $\mathcal{C}$, namely those belonging to the finite union $\mathcal{C}_1 \cup \dots \cup \mathcal{C}_k$. Finally, $\mathcal{C}$ covers X . For given $x \in X$, let m be the smallest integer such that $x \in \mathbb{B}_m$. Then by construction, x belongs to some element of $\mathcal{C}_m$. Thus $\mathcal{C}$ is a locally finite open refinement of $\mathcal{A}$, and $\mathbb{R}^n$ is paracompact.

1.2 Smooth manifolds

Definition 1.23. A Hausdorff paracompact topological space M is called a *topological n -dimensional manifold* if it has the following structure:

1. $\mathfrak{U} = \{U_\alpha\}_{\alpha \in I}$ is an open cover of M .
2. For each $\alpha \in I$, there is a homeomorphism $\varphi_\alpha: U_\alpha \rightarrow \varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$.

The collection of $\{\varphi: U_\alpha \rightarrow \mathbb{R}^n\}$ is called an *atlas*, and each element in an atlas is called a *chart*.

Remark 1.24. Roughly speaking, a topological manifold is a topological space which locally "looks like" a Euclidean space. However, we need to put extra conditions, such as Hausdorffness and paracompactness, to avoid weird cases.

- (1) There exists a topological space which admits an open cover such that each open subset in that open cover is homeomorphic to a Euclidean space, but not Hausdorff (Example 1.21).
- (2) The paracompactness guarantees the existence of the partition of the unity, which is a useful tool.

Definition 1.25. A topological manifold is called a *smooth manifold*, if there exists an atlas $\{\varphi: U_\alpha \rightarrow \mathbb{R}^n\}$ such that for all U_α, U_β , the transition functions:

$$\varphi_\alpha \circ \varphi_\beta^{-1}: \varphi_\beta(U_\alpha \cap U_\beta) \rightarrow \varphi_\alpha(U_\alpha \cap U_\beta)$$

are smooth maps between $\mathbb{R}^n$, and such an atlas is called a *smooth structure*.

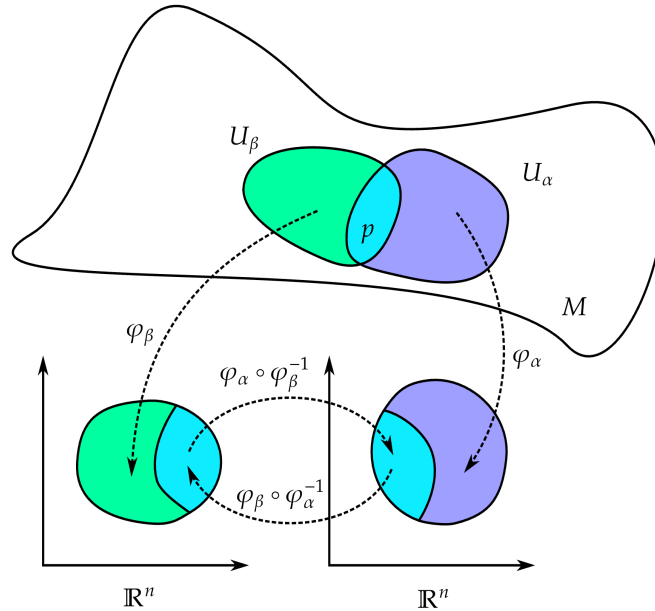


Figure 2: Charts and transition functions

Example 1.26. Define the n -sphere:

$$\mathbb{S}^n = \{x = (x^0, \dots, x^n) \in \mathbb{R}^{n+1} : \|x\|^2 = 1\}.$$

Let $U^+ = \mathbb{S}^n \setminus \{\text{north pole}\}$ and $U^- = \mathbb{S}^n \setminus \{\text{south pole}\}$. Define an atlas:

$$\begin{aligned}\varphi^+ : U^+ &\rightarrow \mathbb{R}^n, & (x^0, \dots, x^n) &\mapsto \frac{1}{1-x^0}(x^1, \dots, x^n), \\ \varphi^- : U^- &\rightarrow \mathbb{R}^n, & (x^0, \dots, x^n) &\mapsto \frac{1}{1+x^0}(x^1, \dots, x^n).\end{aligned}$$

These maps are called *stereographic projections*. The inverse maps are:

$$\begin{aligned}(\varphi^+)^{-1}(x^1, \dots, x^n) &= \frac{1}{1+\|x\|^2} (\|x\|^2 - 1, 2x^1, \dots, 2x^n). \\ (\varphi^-)^{-1}(x^1, \dots, x^n) &= \frac{1}{1+\|x\|^2} (1 - \|x\|^2, 2x^1, \dots, 2x^n).\end{aligned}$$

The transition functions are given by:

$$\varphi^+ \circ (\varphi^-)^{-1}(x) = \frac{x}{\|x\|^2}, \quad \varphi^- \circ (\varphi^+)^{-1}(x) = \frac{x}{\|x\|^2}.$$

The n -sphere is a compact manifold.

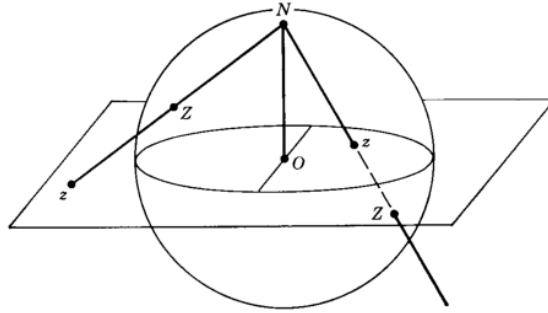


Figure 3: Stereographic projection from north pole

Example 1.27. Let $\mathbb{T}^2 = \{(x, y, z, w) \in \mathbb{R}^4 \mid x^2 + y^2 = 1, z^2 + w^2 = 1\}$. $\mathbb{T}^2$ is a subspace of $\mathbb{R}^4$ with subspace topology. The product of two compact Hausdorff spaces is compact and Hausdorff. Each $\mathbb{S}^1$ has an atlas with two smooth charts. Construct an atlas for $\mathbb{T}^2$ as the product atlas:

$$U_{ij} = U_i \times V_j, \quad \psi_{ij} = \psi_i \times \phi_j : U_{ij} \rightarrow \mathbb{R}^2 \quad (i, j = 1, 2),$$

where ψ_i, ϕ_j are charts of $\mathbb{S}^1$. The transition functions are products of smooth functions, hence smooth.

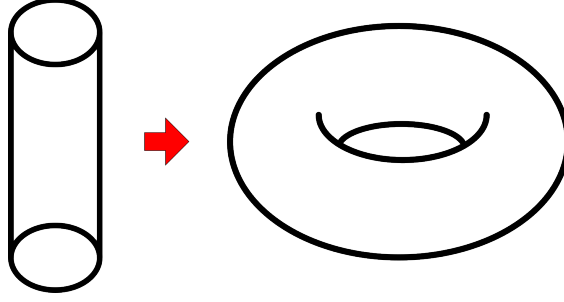


Figure 4: 2-dimensional torus $\mathbb{T}^2$.

Definition 1.28. Let M be an n -dimensional manifold. A subset $M' \subset M$ is called a *submanifold* if it satisfies: For each $q \in M'$, there exists a local coordinate system $(U; x^1, \dots, x^n)$ such that:

$$M' \cap U = \{q \in U \mid x^{k+1}(q) = \dots = x^n(q) = 0\}.$$

Then M' is k -dimensional. If $\dim(M') = n - k$, then M' is called a submanifold of codimension k in M .

Example 1.29.

$$\mathbb{S}^k \subset \mathbb{S}^n = \{x = (x^0, \dots, x^k, 0, \dots, 0) \in \mathbb{R}^{n+1} : \|x\|^2 = 1\}$$

is a submanifold of $\mathbb{S}^n$.

Definition 1.30. Let M and N be m - and n -dimensional manifolds respectively.

- (1) A continuous map $f: M \rightarrow N$ is called *smooth* at $p \in M$, if for every chart $(U_\alpha, \varphi_\alpha)$ containing p and every chart (V_β, ψ_β) containing $f(p) \in N$, the map:

$$\psi_\beta \circ f \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap f^{-1}(V_\beta)) \rightarrow \psi_\beta(V_\beta)$$

is smooth at $\varphi_\alpha(p)$.

- (2) A continuous map $f: M \rightarrow N$ is called *smooth* if it is smooth at every point $p \in M$.
- (3) A continuous map $f: M \rightarrow N$ is called *diffeomorphism*, if f is smooth, bijective and it has a smooth inverse.

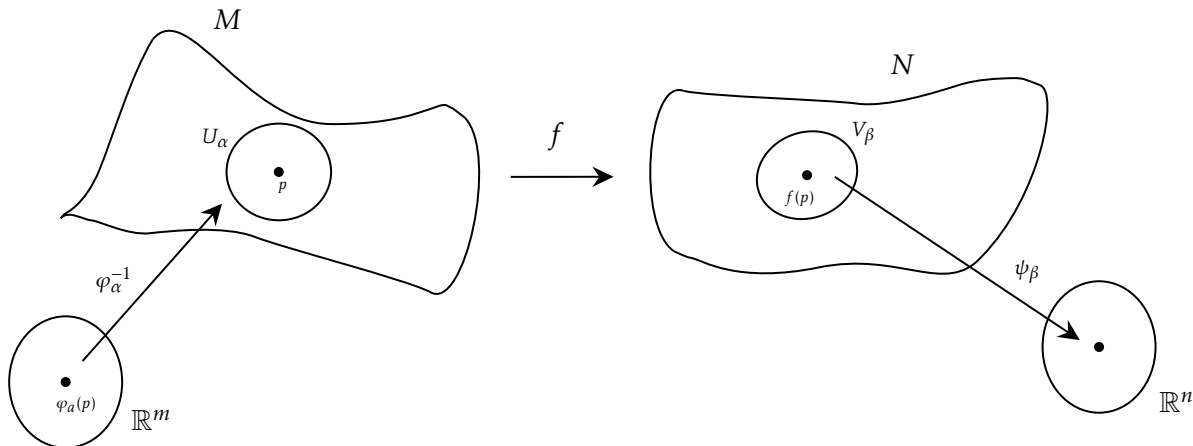


Figure 5: The map $f : M \rightarrow N$ is C^∞ at p .

Example 1.31. Let M and N be smooth manifolds. Then the projection $M \times N \rightarrow N$ is a smooth map.

Example 1.32. A rotation of $\mathbb{S}^1$ is a diffeomorphism.

1.3 Tangent bundle and cotangent bundle

1.3.1 Tangent spaces

Definition 1.33. A *tangent vector* X_p at $p \in M$ is a map $X_p : C^\infty(M) \rightarrow \mathbb{R}$ satisfying:

1. *Linearity*: $X_p(af + bg) = aX_p(f) + bX_p(g)$ for $a, b \in \mathbb{R}$;
2. *Leibniz Rule*: $X_p(fg) = f(p)X_p(g) + X_p(f)g(p)$.

A tangent vector X_p behaves like a derivative. The set of tangent vectors at p forms a vector space:

$$(X_p + Y_p)(f) = X_p(f) + Y_p(f), \quad (aX_p)(f) = a(X_p(f)).$$

This vector space is called the *tangent space* $T_p M$ at $p \in M$.

Definition 1.34. An *integral curve* of a vector field $X \in \mathfrak{X}(M)$ is a smooth map $\gamma : I \rightarrow M$ (where $I \subset \mathbb{R}$ is an open interval) such that $\dot{\gamma}(t) = X_{\gamma(t)}$.

Example 1.35. For $X = \alpha \frac{\partial}{\partial x} + \beta \frac{\partial}{\partial y}$ in $\mathbb{R}^2$, the integral curve is a translation:

$$\gamma_t(x, y) = (x + t\alpha, y + t\beta).$$

Given local coordinates $(x^1, \dots, x^n)$, the integral curve satisfies:

$$\frac{d}{dt} x^i = \alpha^i(x^1(t), \dots, x^n(t)).$$

These are ODEs. By the existence theorem, there is a unique solution $\gamma : (-\epsilon, \epsilon) \rightarrow M$ with $\gamma(0) = p$ for sufficiently small $\epsilon > 0$.

Theorem 1.36. For $X \in \mathfrak{X}(M)$ and $p \in M$, there exists an open interval $I \subset \mathbb{R}$ containing 0 and an integral curve $\gamma: I \rightarrow M$ for X with $\gamma(0) = p$. If $\tilde{\gamma}: \tilde{I} \rightarrow M$ is another integral curve for X with $\tilde{\gamma}(0) = p$, then $\gamma = \tilde{\gamma}$ on $I \cap \tilde{I}$.

Definition 1.37. If $\gamma(t)$ is defined for all $t \in \mathbb{R}$, then X is *complete*. If M is compact, every vector field is complete. In this case, the flow is a smooth map:

$$\Phi: \mathbb{R} \times M \rightarrow M, \quad (t, p) \mapsto \Phi_t(p)$$

satisfying:

$$\Phi_{t+s}(p) = \Phi_t(\Phi_s(p)), \quad \frac{d}{dt}\Phi_t(p) = X_{\Phi_t(p)}.$$

There is another way to think about tangent vectors. A curve starting from p is a smooth map $\gamma: I \rightarrow M$ with $\gamma(0) = p$ where $I \subset \mathbb{R}$ is a non-empty open interval. Two curves γ_1, γ_2 are *tangent* at p if:

$$\gamma_1(0) = p = \gamma_2(0), \quad \left. \frac{d}{dt} \right|_{t=0} \varphi(\gamma_1(t)) = \left. \frac{d}{dt} \right|_{t=0} \varphi(\gamma_2(t))$$

for any chart (U, φ) about p . This gives an equivalent relation, and the set of all curves starting from p up to this equivalence classes forms $T_p M$. For $f \in C^\infty(M)$, we can take the derivative of f along γ :

$$X_p(f) = \left. \frac{d}{dt} \right|_{t=0} f(\gamma(t)),$$

where $X_p = \left. \frac{d}{dt} \right|_{t=0} \psi(\gamma(t))$ and φ is any chart about p .

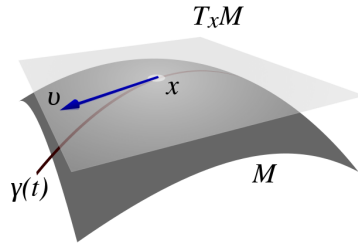


Figure 6: Tangent vector at $x \in M$

Given local coordinates $\varphi = (x^1, \dots, x^n)$, the tangent vector along a curve γ can be written as:

$$X_p = \sum_{i=1}^n X_p^i \frac{\partial}{\partial x^i} \Big|_p \quad \text{where } dx_p^i = \left. \frac{d}{dt} x^i(\gamma(t)) \right|_{t=0}$$

Thus $(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n})$ form a basis for $T_p M$. For another coordinate system $(y^1, \dots, y^n)$:

$$\frac{\partial}{\partial y^j} \Big|_p = \sum_{i=1}^n \frac{\partial x^i}{\partial y^j}(p) \frac{\partial}{\partial x^i} \Big|_p,$$

where $\frac{\partial x^i}{\partial y^j}(p)$ is the Jacobian at p .

Definition 1.38. Let $f: M \rightarrow N$ be a smooth map between manifolds. For $p \in M$ and each $v \in T_p M$, choose a curve $\gamma: (-\epsilon, \epsilon) \rightarrow M$ with $\gamma(0) = p$ and $\gamma'(0) = v$. The *tangent map* of f at p is:

$$(df)_p: T_p M \rightarrow T_{f(p)} N$$

defined by:

$$((df)_p)(v) = \beta'(0),$$

where $\beta = f \circ \gamma: (-\epsilon, \epsilon) \rightarrow N$.

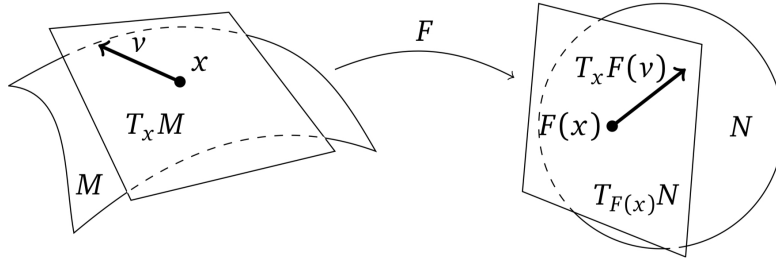


Figure 7: Tangent map at $x \in M$ such that $(dF)_x: T_x M \rightarrow T_{F(x)} N$.

Proposition 1.39. Let M, N, P be manifolds, $f: M \rightarrow N$ and $g: N \rightarrow P$ smooth maps. For $q \in M$:

$$(dg \circ f)_q = dg_{f(q)} \circ (df)_q: T_q M \rightarrow T_{g(f(q))} P.$$

Definition 1.40. Let M, N be smooth manifolds.

1. A map $f: M \rightarrow N$ is a *diffeomorphism* if it is smooth, bijective, and f^{-1} is smooth.
2. $f: M \rightarrow N$ is a *local diffeomorphism* at $p \in M$ if there exist neighborhoods $U \ni p$ and $V \ni f(p)$ such that $f: U \rightarrow V$ is a diffeomorphism.

Example 1.41. Consider $f: \mathbb{R} \rightarrow \mathbb{S}^1$, $f(\theta) = e^{\sqrt{-1}\theta}$. This is a local diffeomorphism but not a diffeomorphism.

Lemma 1.42. Every bijective local diffeomorphism is a diffeomorphism.

Definition 1.43. Let $f: M \rightarrow N$ be a smooth map. The *rank* of f at p is defined as the rank of the tangent map:

$$\text{rank}((df)_p: T_p M \rightarrow T_{f(p)} N).$$

Definition 1.44. Let $f: M \rightarrow N$ be a smooth map and $(df)_p: T_p M \rightarrow T_{f(p)} N$ the tangent map.

1. f is an *immersion* if $(df)_p$ is injective for all $p \in M$;
2. f is a *submersion* if $(df)_p$ is surjective for all $p \in M$;
3. f is an *embedding* if it is an immersion and $f: M \rightarrow f(M)$ is a homeomorphism;
4. If $M \subset N$ and the inclusion map $i: M \rightarrow N$ is an embedding, then M is a *submanifold* of N .

Remark 1.45.

- An immersion may not be injective (e.g., $f: \mathbb{R} \rightarrow \mathbb{S}^1$, $f(\theta) = e^{\sqrt{-1}\theta}$);
- A submersion may not be surjective (e.g., $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \arctan(x)$).

1.3.2 Tangent Bundle

Consider the collection of tangent spaces over each point of a manifold:

$$TM = \bigcup_{p \in M} T_p M = \{(p, v_p) \mid p \in M, v_p \in T_p M\}.$$

There is a natural projection $\pi: TM \rightarrow M$ sending $v_p \in T_p M$ to p .

Theorem 1.46. Let M be a n -dimensional smooth manifold. Then the tangent bundle TM is a $2n$ -dimensional manifold called the tangent bundle, and $\pi: TM \rightarrow M$ is a smooth map.

Proof. Given local coordinates $(x^1, \dots, x^n)$ on a chart (U, φ) , for each $p \in U$ and $v_p \in T_p M$, there are $\alpha^1, \dots, \alpha^n \in \mathbb{R}$ such that:

$$v_p = \sum_{i=1}^n \alpha^i \frac{\partial}{\partial x^i} \Big|_p.$$

This gives a bijection:

$$\tilde{\varphi}: \pi^{-1}(U) \rightarrow \varphi(U) \times \mathbb{R}^n, \quad v_p \mapsto (x^1(p), \dots, x^n(p), \alpha^1, \dots, \alpha^n).$$

The transition functions between charts are smooth, so TM is a manifold. □

Definition 1.47. A smooth map $X: M \rightarrow TM$ is called a *section* of $\pi: TM \rightarrow M$ if $\pi \circ X = \text{id}_M$. It is called a *vector field*, denoted $\mathfrak{X}(M) = C^\infty(M, TM)$.

Lemma 1.48. A vector field $X \in \mathfrak{X}(M)$ gives a map $C^\infty(M) \rightarrow C^\infty(M)$ satisfying:

1. $X(af + bg) = aX(f) + bX(g)$, where $a, b \in \mathbb{R}$;
2. $X(fg) = gX(f) + fX(g)$.

Example 1.49. On $\mathbb{S}^n \subset \mathbb{R}^{n+1}$, we have a vector field:

$$X = \sum_{i=0}^n X^i \frac{\partial}{\partial x^i},$$

where

$$X^i = \begin{cases} (-x^1, x^0, -x^3, x^2, \dots, -x^{2k+1}, x^{2k}) & n = 2k + 1, \\ (-x^1, x^0, -x^3, x^2, \dots, -x^{2k+1}, x^{2k}, 0) & n = 2k + 2. \end{cases}$$

Definition 1.50. For $X, Y \in \mathfrak{X}(M)$ and $f \in C^\infty(M)$, $[X, Y] = XY - YX$ is a vector field called the *Lie bracket*. It satisfies:

1. Bilinearity;
2. Antisymmetry: $[X, Y] = -[Y, X]$;
3. Jacobi identity: $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$.

1.3.3 Orientability

There are several equivalent definitions for orientability. Let M be an n -dimensional manifold.

Definition 1.51. An *orientation* on M is an equivalence class of oriented atlases. An atlas $\{U_\alpha, \varphi_\alpha\}$ is oriented if for every pair of charts U_α, φ_α and U_β, φ_β , the transition function:

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

has positive Jacobian determinant:

$$\det(D(\varphi_\beta \circ \varphi_\alpha^{-1})(x)) > 0.$$

Definition 1.52. At each point $p \in M$, choose one of the two equivalence classes of ordered bases of $T_p M$, where two bases are equivalent if the change-of-basis matrix has positive determinant. This choice must vary smoothly over M .

Example 1.53.

- Sphere and torus are orientable.
- Möbius strip and Klein bottle are not orientable.
- Real projective space $\mathbb{RP}^n$ is orientable if and only if n is odd.

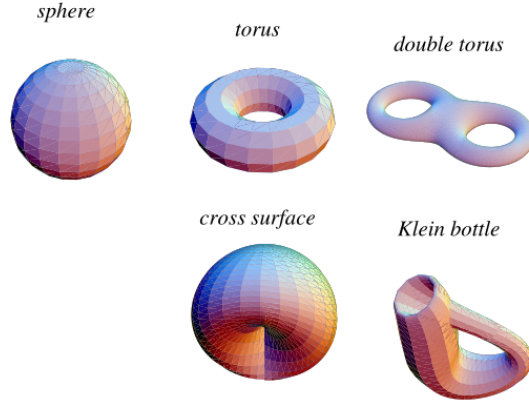


Figure 8: Examples of orientable and non-orientable compact manifolds.

1.3.4 Cotangent bundle

Definition 1.54. The dual space T_p^*M of the tangent space T_pM is called the *cotangent space*. For local coordinates $(x^1, \dots, x^n)$, the basis $(dx^1|_p, \dots, dx^n|_p)$ of T_p^*M satisfies:

$$dx^i|_p \left(\frac{\partial}{\partial x^j} \Big|_p \right) = \delta_j^i.$$

The *cotangent bundle* is:

$$T^*M = \bigcup_{p \in M} T_p^*M$$

and sections of T^*M are called *differential 1-forms*, denoted $\Omega^1(M) = C^\infty(M, T^*M)$.

It is natural to consider the algebra generated by the basis of T_p^*M over $\mathbb{R}$ satisfying $dx^i \wedge dx^j = -dx^j \wedge dx^i$. The product $\wedge$ is called the *wedge product*.

Definition 1.55. For an n -manifold, we define the *exterior algebra* as follows

$$\Lambda^* T_p^*M = \bigoplus_{k=0}^n \Lambda^k T_p^*M.$$

An element $\omega \in \Lambda^k T_p^*M$ defines an alternating k -linear map $T_pM \times \dots \times T_pM \rightarrow \mathbb{R}$. The bundle

$$\Lambda^k T^*M = \bigcup_p \Lambda^k T_p^*M$$

is a smooth manifold. Its sections are called *differential k -forms*, denoted $\Omega^k(M) = C^\infty(M, \Lambda^k T^*M)$.

Remark 1.56. The notation $\Lambda^* T_p^* M$ denotes the *exterior algebra* of the cotangent space $T_p^* M$. It is defined as the direct sum of all exterior powers $\Lambda^* T_p^* M = \bigoplus_{k=0}^n \Lambda^k T_p^* M$, where $\Lambda^k T_p^* M$ is the space of alternating k -linear forms on $T_p M$, with $\Lambda^0 T_p^* M \cong \mathbb{R}$ and $\Lambda^1 T_p^* M = T_p^* M$. Thus, the exterior algebra collects all differential forms of every degree at the point p into a single graded vector space.

Definition 1.57. M is *orientable* if and only if there exists a nowhere-vanishing differential n -form, i.e., $\omega \in \Omega^n(M)$. Such ω defines an orientation: a basis $(v_1, \dots, v_n)$ of $T_p M$ is positively oriented if:

$$\omega_p(v_1, \dots, v_n) > 0.$$

Definition 1.58. There is a unique linear map $d: \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ called the *exterior derivative* satisfying:

1. For $f \in C^\infty(M)$, df is the usual differential;
2. $d \circ d = 0$;
3. $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$ for $\omega \in \Omega^k(M)$.

In local coordinates:

$$d \left(\sum_{i_1 < \dots < i_k} \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} \right) = \sum_{i_1 < \dots < i_k} d\omega_{i_1 \dots i_k} \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

Definition 1.59. For a smooth map $f: M \rightarrow N$, the *pullback* $f^*: \Omega^*(N) \rightarrow \Omega^*(M)$ is defined by:

$$(f^* \omega)(X_1, \dots, X_k) = \omega(df(X_1), \dots, df(X_k)).$$

It satisfies:

1. Linearity;
2. $f^*(\omega \wedge \eta) = f^* \omega \wedge f^* \eta$;
3. $(g \circ f)^* = f^* \circ g^*$;
4. Commutes with d : $df^* = f^* d$.

1.4 De Rham cohomology

1.4.1 De Rham complex

Let us now consider a geometric example in calculus. Let $U \subset \mathbb{R}^n$ be an open set, and $\{x^1, \dots, x^n\}$ be its coordinates. Introduce

$$\Omega^0(U) = \{\text{smooth functions on } U\} = C^\infty(U)$$

$$\Omega^1(U) = \{\text{smooth 1-forms on } U\}.$$

An element $\alpha \in \Omega^1(U)$ can be written as

$$\alpha = \sum_{i=1}^n \alpha_i(x) dx^i \in \Omega^1(U), \quad \alpha_i \in C^\infty(U).$$

The total differential gives a map

$$\begin{aligned} \Omega^0(U) &\xrightarrow{d} \Omega^1(U) \\ f &\mapsto df = \sum_i \partial_i f dx^i, \end{aligned}$$

which can be extended to a cochain complex

$$\Omega^0(U) \xrightarrow{d} \Omega^1(U) \xrightarrow{d} \Omega^2(U) \rightarrow \dots \xrightarrow{d} \Omega^n(U),$$

where $\Omega^p(U) = \{\text{smooth } p\text{-forms on } U\}$. An element $\xi \in \Omega^p(U)$ takes the form

$$\xi = \sum_{i_1, \dots, i_p} \xi_{i_1 \dots i_p}(x) dx^{i_1} \wedge \dots \wedge dx^{i_p}$$

with $\xi_{i_1 \dots i_p}(x) \in C^\infty(U)$ and the skew-symmetry property

$$dx^i \wedge dx^j = -dx^j \wedge dx^i, \quad \forall i, j$$

In particular, $dx^i \wedge dx^i = 0$. Thus we can also write

$$\Omega^p(U) = \bigoplus_{1 \leq i_1 < \dots < i_p \leq n} C^\infty(U) dx^{i_1} \wedge \dots \wedge dx^{i_p}.$$

The de Rham differential $d: \Omega^p(U) \rightarrow \Omega^{p+1}(U)$ is defined by

$$d(\xi) = \sum_{i_1, \dots, i_p, k} \partial_{x^k} \xi_{i_1 \dots i_p} dx^k \wedge dx^{i_1} \wedge \dots \wedge dx^{i_p}.$$

Thus

$$\Omega^0(U) \xrightarrow{d} \Omega^1(U) \xrightarrow{d} \Omega^2(U) \rightarrow \dots \rightarrow \Omega^n(U)$$

defines a cochain complex called the de Rham complex and denoted by $(\Omega^*(U), d)$.

Definition 1.60. A p -form $\alpha \in \Omega^p$ is called

- *closed* if $d\alpha = 0$

- *exact* if $\alpha = d\beta$ for some $p-1$ form β .

The quotient space

$$H_{dR}^p(U) = H^p(\Omega^*, d) = \frac{\text{closed } p\text{-forms}}{\text{exact } p\text{-forms}}$$

is called the p -th *de Rham cohomology*.

One important property is that the de Rham differential d is intrinsic: it does not depend on the choice of coordinates. Thus, we can define $(\Omega^*(M), d)$ on a smooth manifold M by gluing the above local construction. Then the de Rham cohomology on manifold $H^p(M) = H_{dR}^p(M)$ is well-defined. We can also consider $(\Omega_c^*(M), d)$, where

$$\Omega_c^*(M) = \{p\text{-form on } M \text{ with compact support}\}.$$

Then we can define de Rham cohomology with compact support

$$H_c^p(M) = H^p(\Omega_c^*(M), d).$$

Remark 1.61. Consider a smooth manifold M .

- The zeroth cohomology group $H^0(M)$ consists of closed 0-forms modulo exact 0-forms. Since every 0-form is just a smooth function and $df = 0$ means f is constant on each connected component, we have $H^0(M) \cong \mathbb{R}^{\#\pi_0(M)}$, where $\pi_0(M)$ denotes the set of connected components of M . Thus H^0 measures the number of connected components.
- The first cohomology group $H^1(M)$ consists of closed 1-forms modulo exact 1-forms. It detects the existence of nontrivial loops in M . For example, on the circle $M = \mathbb{S}^1$ we have $H^1(\mathbb{S}^1) \cong \mathbb{R}$, corresponding to the integral of a closed 1-form around the fundamental loop.
- The second cohomology group $H^2(M)$ classifies closed 2-forms modulo exact 2-forms. On an oriented surface M of genus g , such as the 2-torus $\mathbb{T}^2$, one finds $H^2(\mathbb{T}^2) \cong \mathbb{R}$, generated by the area form. In general, H^2 captures information about “fluxes” or surface integrals, such as the total curvature or electromagnetic flux through a surface.

Example 1.62. Cohomology groups for $\mathbb{S}^1, \mathbb{S}^2, \mathbb{T}^2$.

- H^0 always gives $\mathbb{R}$ because each space is connected.
- H^1 measures independent loops:
 - $\mathbb{S}^1$: one loop $\Rightarrow H^1 \cong \mathbb{R}$.
 - $\mathbb{S}^2$: no nontrivial loops $\Rightarrow H^1 = 0$.
 - $\mathbb{T}^2$: two independent loops $\Rightarrow H^1 \cong \mathbb{R}^2$.

- H^2 measures closed 2-forms (“fluxes”):
 - $\mathbb{S}^1$: no 2-dimensional surfaces $\Rightarrow H^2 = 0$.
 - $\mathbb{S}^2$: oriented area form generates $H^2 \cong \mathbb{R}$.
 - $\mathbb{T}^2$: oriented area form generates $H^2 \cong \mathbb{R}$.

1.5 Integration of differential forms

1.5.1 Integral Theorems

Let M be an oriented smooth n -manifold and $\omega \in \Omega^n(M)$ has compact support. To define $\int_M \omega$, we need partitions of unity.

Theorem 1.63. *Every open cover has a locally finite refinement.*

Theorem 1.64. *A partition of unity subordinate to a locally finite open cover $\{U_\alpha\}$ is a collection $\{\rho_\alpha\} \subset C^\infty(M, [0, 1])$ such that:*

1. $\text{supp}(\rho_\alpha) \subset U_\alpha$;
2. $\sum_\alpha \rho_\alpha = 1$.

In local coordinates, $\rho_\alpha \omega = f_\alpha(x) dx^1 \wedge \cdots \wedge dx^n$. Define:

$$\int_M \rho_\alpha \omega = \int \cdots \int f_\alpha(x) dx^1 \cdots dx^n.$$

Then define:

$$\int_M \omega = \sum_\alpha \int_M \rho_\alpha \omega.$$

This is independent of the choice of cover and partition of unity, and it is a finite summation, as ω has compact support.

Theorem 1.65. *Let M be an oriented n -manifold with boundary. If $\omega \in \Omega^{n-1}(M)$ has compact support, then:*

$$\int_M d\omega = \int_{\partial M} \omega$$

Corollary 1.66. *If M is compact and oriented without boundary, then for any $\omega \in \Omega^{n-1}(M)$:*

$$\int_M d\omega = 0.$$

Corollary 1.67.

- Fundamental Theorem of Calculus:

$$\int_a^b f'(x) dx = f(b) - f(a).$$

- Fundamental Theorem for Line Integrals:

$$\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A).$$

- Green's Theorem:

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_{\partial D} P dx + Q dy.$$

- Stokes' Theorem (Vector Calculus):

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}.$$

- Divergence Theorem:

$$\iiint_V (\nabla \cdot \mathbf{F}) dV = \iint_{\partial V} \mathbf{F} \cdot d\mathbf{S}.$$

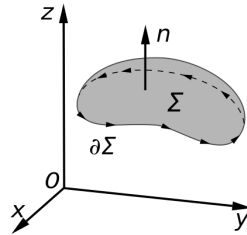


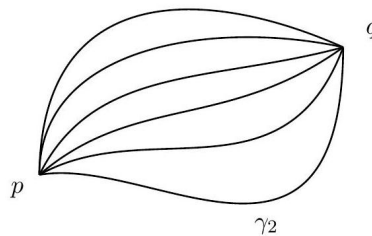
Figure 9: Stokes' theorem with surface Σ its boundary $\partial\Sigma$ and the normal vector n .

Example 1.68. Let $\alpha \in \Omega^1(\mathbb{R}^n)$ and $d\alpha = 0$. Let $p, q \in \mathbb{R}^n$ be two points. Define

$$\int_p^q \alpha = \int_\gamma \alpha$$

by choosing any smooth path γ connecting p and q . It does not depend on the choice of γ since any such path γ_1 can be continuously deformed to another γ_2 . Then

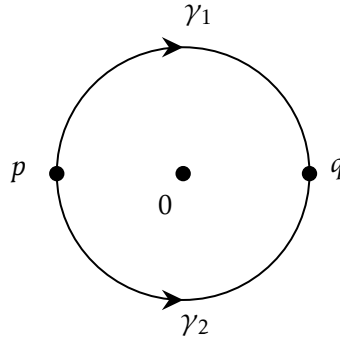
$$\int_{\gamma_1} \alpha = \int_{\gamma_2} \alpha \quad \text{by Stokes' Theorem.}$$



Example 1.69. Consider $\mathcal{U} = \mathbb{R}^2 - \{0\}$ and

$$\alpha = \frac{x dy - y dx}{x^2 + y^2}, \quad d\alpha = 0.$$

We consider two paths from $p = -1$ to $q = 1$ by following the upper and lower semicircle.



Since there is a hole in the middle, γ_1 can not be deformed to γ_2 inside the region $\mathcal{U}$. This can be also seen by computing

$$\int_{\gamma_1} \alpha - \int_{\gamma_2} \alpha = \oint_C \alpha$$

In terms of polar coordinates $x = r \cos \theta, y = r \sin \theta$,

$$\alpha = d\theta, \quad \oint_C \alpha = \int_0^{2\pi} d\theta = 2\pi \neq 0.$$

1.5.2 Cohomology and Integration

Theorem 1.70 (Poincaré lemma).

$$H^p(\mathbb{R}^n) = \begin{cases} \mathbb{R}, & p = 0 \\ 0, & p > 0 \end{cases} \quad H_c^p(\mathbb{R}^n) = \begin{cases} \mathbb{R}, & p = n \\ 0, & p \neq n \end{cases}$$

A generator of $H^0(\mathbb{R}^n)$ is the constant function $f \equiv 1$, and a generator of $H_c^n(\mathbb{R}^n)$ is a compactly supported form $\alpha = f(x) dx^1 \wedge \cdots \wedge dx^n$, where $f \in C_c^\infty(\mathbb{R}^n)$, such that $\int_{\mathbb{R}^n} \alpha = 1$.

Observe that any n -form $\alpha \in \Omega_c^n(\mathbb{R}^n)$ is closed. On the other hand, if $\alpha = d\beta$ for $\beta \in \Omega_c^{n-1}(\mathbb{R}^n)$, then

$$\int_{\mathbb{R}^n} d\beta = \int_{\mathbb{B}} d\beta = \int_{\partial\mathbb{B}} \beta = 0$$

where $\mathbb{B}$ is a large enough ball. The integration map

$$\begin{aligned} \int_{\mathbb{R}^n} : H_c^n(\mathbb{R}^n) &\longrightarrow \mathbb{R} \\ \alpha &\longmapsto \int_{\mathbb{R}^n} \alpha \end{aligned}$$

is the isomorphism described in the Poincaré lemma. Viewed differently, the map

$$\begin{aligned} \Omega_c^n(\mathbb{R}^n) &\longrightarrow H_c^n(\mathbb{R}^n) \stackrel{\text{Poincaré}}{\cong} \mathbb{R} \\ \alpha &\longmapsto [\alpha] \end{aligned}$$

sending a top n -form to its cohomology class determines precisely the integration (up to a rescaling constant). Thus

$$\int_{\mathbb{R}^n} \Longleftrightarrow H_c^n$$

i.e. "integration" can be modeled by a homological operation.

Part II

Riemannian Geometry

2 Riemannian manifold and Levi-Civita connection

2.1 Vector bundle

Definition 2.1. Let M be a smooth manifold. A (real) vector bundle E of rank r on M consists of the following data:

- (1) E is a smooth manifold with surjective map $\pi: E \rightarrow M$, such that
 - (1) For all $x \in M$, fibre E_x is a $\mathbb{R}$ -vector space of dimension r .
 - (2) For all $x \in M$, there exists $x \in U \subseteq M$ and there is a diffeomorphism $\varphi: \pi^{-1}(U) \rightarrow U \times \mathbb{R}^r$ such that

$$\begin{array}{ccc}
 \pi^{-1}(U) & \xrightarrow{\pi} & U \\
 \searrow \varphi & \curvearrowright p_1 & \nearrow \\
 & U \times \mathbb{R}^r & \xrightarrow{p_2} \mathbb{R}^r
 \end{array}$$

and for all $y \in U$, $E_y \xrightarrow{p_2 \circ \varphi} \mathbb{R}^r$ is a $\mathbb{R}$ -vector space isomorphism. The pair (U, φ) is called a trivialization of E over U .

Remark 2.2 (transition functions). Let $(U_\alpha, \varphi_\alpha), (U_\beta, \varphi_\beta)$ be local trivializations. Then $\varphi_\alpha \circ \varphi_\beta^{-1}: (U_\alpha \cap U_\beta) \times \mathbb{R}^r \rightarrow (U_\alpha \cap U_\beta) \times \mathbb{R}^r$ induces a smooth map

$$g_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow \text{GL}(r, \mathbb{R}),$$

where $g_{\alpha\beta}$ is called transition function. Furthermore, it satisfies

$$\begin{aligned}
 g_{\alpha\beta} g_{\beta\gamma} g_{\gamma\alpha} &= \text{id} \quad \text{on } U_\alpha \cap U_\beta \cap U_\gamma \\
 g_{\alpha\alpha} &= \text{id} \quad \text{on } U_\alpha.
 \end{aligned}$$

In fact, transition functions contain all information about vector bundle since a vector bundle is locally trivial, so how are these trivial pieces glued together really matters.

Definition 2.3. Let M be a smooth manifold. A (real) vector bundle E of rank r on M consists of the following data:

- (1) open covering $\{U_\alpha\}$ of M .
- (2) smooth functions $\{g_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow \text{GL}(r, \mathbb{R})\}$ satisfies

$$\begin{aligned}
 g_{\alpha\beta} g_{\beta\gamma} g_{\gamma\alpha} &= \text{id} \quad \text{on } U_\alpha \cap U_\beta \cap U_\gamma \\
 g_{\alpha\alpha} &= \text{id} \quad \text{on } U_\alpha.
 \end{aligned}$$

Definition 2.4. A vector bundle of rank one is called a *line bundle*.

Example 2.5. Let M be a smooth manifold. Then $M \times \mathbb{R}^r$ is called *trivial vector bundle* of rank r on M .

Definition 2.6. Let M be a smooth manifold and $\pi: E \rightarrow M$ be a vector bundle. A subset $F \subseteq E$ is called a *subbundle* of rank s , if

- (1) For all $x \in M$, $F \cap E_x$ is a subspace of E_x with dimension s .
- (2) $\pi|_F: F \rightarrow M$ is a vector bundle.

Definition 2.7. Let M be a smooth manifold and $\pi: E \rightarrow M$ be a vector bundle. A (*smooth*) *section* of E is a smooth map $s: M \rightarrow E$ such that $\pi \circ s = \text{id}_M$.

Definition 2.8. A section of $\bigotimes^s T^*M \otimes \bigotimes^r TM$ is called an (r, s) -*tensor*.

Example 2.9. A smooth function f is an $(0, 0)$ -tensor.

Example 2.10. A vector field X is an $(0, 1)$ -tensor.

Example 2.11. A 1-form ω is an $(1, 0)$ -tensor.

2.2 Riemannian manifold

2.2.1 Riemannian metric

Definition 2.12.

- (1) A *Riemannian metric* is an $(2, 0)$ -tensor such that for any $x \in M$, $g_x: T_x M \times T_x M \rightarrow \mathbb{R}$ is a positive definite symmetric bilinear form.
- (2) A *Riemannian manifold* (M, g) is a smooth manifold M together with a Riemannian metric g .

Example 2.13. Let $\mathbb{S}^n(R)$ denote n -dimensional sphere with radius R . There is a natural inclusion $f: \mathbb{S}^n(R) \hookrightarrow (\mathbb{R}^{n+1}, g_{\text{can}})$, and we can use f to pullback g_{can} to obtain a metric on $\mathbb{S}^n(R)$, denoted by g . Given a local chart (U, φ, x^i) , we can write

$$f(x^1, \dots, x^n) = \left(x^1, \dots, x^n, \sqrt{R^2 - \sum_{i=1}^n (x^i)^2} \right).$$

For any $\frac{\partial}{\partial x^i}$, we have

$$df \left(\frac{\partial}{\partial x^i} \right) = \frac{\partial f^j}{\partial x^i} \frac{\partial}{\partial x^j} = \frac{\partial}{\partial x^i} - \frac{x^i}{\sqrt{R^2 - \sum_{i=1}^n (x^i)^2}} \frac{\partial}{\partial x^{n+1}}.$$

Thus for any two $\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}$ we have

$$\begin{aligned} g\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right) &= g_{\text{can}}\left(df \frac{\partial}{\partial x^i}, df \frac{\partial}{\partial x^j}\right) \\ &= g_{\text{can}}\left(\frac{\partial}{\partial x^i} - \frac{x^i}{\sqrt{R^2 - \sum_{i=1}^n (x^i)^2}} \frac{\partial}{\partial x^{n+1}}, \frac{\partial}{\partial x^j} - \frac{x^j}{\sqrt{R^2 - \sum_{i=1}^n (x^i)^2}} \frac{\partial}{\partial x^{n+1}}\right) \\ &= \delta_{ij} + \frac{x^i x^j}{R^2 - \sum_{i=1}^n (x^i)^2}, \end{aligned}$$

which implies

$$g_{ij} = \delta_{ij} + \frac{x^i x^j}{T^2},$$

where $T^2 = R^2 - \sum_{i=1}^n (x^i)^2$.

Example 2.14. Let $\mathbb{H}^n(R) = \{(x^1, \dots, x^{n-1}, y) \in \mathbb{R}^n \mid y > 0\}$ with Riemannian metric

$$g = R^2 \frac{\delta_{ij} dx^i \otimes dx^j + dy \otimes dy}{y^2}.$$

It is called *hyperbolic upper plane*.

Example 2.15. Let $\mathbb{B}^n(R) = \{x \in \mathbb{R}^n \mid |x| < R\}$ with Riemannian metric

$$g = 4R^4 \frac{\delta_{ij} dx^i \otimes dx^j}{(R^2 - |x|^2)^2}.$$

It is called *Poincaré disk*.

Definition 2.16. A Riemannian metric $\langle \cdot, \cdot \rangle$ on a Lie group G is called *left-invariant* if

$$\langle (dL_g)X, (dL_g)Y \rangle = \langle X, Y \rangle$$

holds for arbitrary $g \in G$ and vector fields X, Y .

Remark 2.17. The right-invariant metric can be defined in a similar statement, and a Riemannian metric which is both left-invariant and right-invariant is called a *bi-invariant metric*.

Proposition 2.18. *There is a bijective correspondence between left-invariant metrics on a Lie group G , and inner products on the Lie algebra $\mathfrak{g}$ of G .*

Proof. Let $\langle \cdot, \cdot \rangle_e$ be an inner product on Lie algebra $\mathfrak{g}$. Then we have a Riemannian metric on G defined as follows

$$\langle X_g, Y_g \rangle := \langle (dL_{g^{-1}})X_g, (dL_{g^{-1}})Y_g \rangle_e,$$

where X, Y are two vector fields on G . It is left-invariant, since

$$\begin{aligned} \langle (dL_h)X_g, (dL_h)Y_g \rangle &= \langle (dL_{(hg)^{-1}}) \circ (dL_h)X_g, (dL_{(hg)^{-1}}) \circ (dL_h)Y_g \rangle_e \\ &= \langle (dL_{g^{-1}})X_g, (dL_{g^{-1}})Y_g \rangle_e. \end{aligned}$$

Conversely, if we have a left-invariant inner product $\langle \cdot, \cdot \rangle$ on G , then it is clear we have an inner product on $\mathfrak{g}$, by considering its value at identity. Furthermore, these two constructions are inverse to each other. $\square$

Proposition 2.19. *There is a bijective correspondence between bi-invariant metrics on a Lie group G , and Ad-invariant inner products on the Lie algebra $\mathfrak{g}$ of G .*

Proof. Given an Ad-invariant inner product $\langle \cdot, \cdot \rangle_e$ on the Lie algebra $\mathfrak{g}$, by Proposition 2.18, there is a left-invariant metric $\langle \cdot, \cdot \rangle$ on G , it suffices to check it's also right-invariant:

$$\begin{aligned} \langle (dR_h)X_g, (dR_h)Y_g \rangle &= \langle (dL_{(gh)^{-1}}) \circ (dR_h)X_g, (dL_{(gh)^{-1}}) \circ (dR_h)Y_g \rangle_e \\ &= \langle \text{Ad}(h^{-1})(dL_{g^{-1}})X_g, \text{Ad}(h^{-1})(dL_{g^{-1}})Y_g \rangle_e \\ &= \langle (dL_{g^{-1}})X_g, (dL_{g^{-1}})Y_g \rangle_e. \end{aligned}$$

Conversely, if we start with a bi-invariant metric, then its restriction to the Lie algebra is an Ad-invariant metric, since $\text{Ad}(g)$ is exactly the differential of $L_g \circ R_{g^{-1}}$. $\square$

2.2.2 Riemannian isometry

Definition 2.20. A smooth map $f: (M, g_M) \rightarrow (N, g_N)$ between Riemannian manifolds is called a *local isometry* if for each point $p \in M$ the differential $(df)_p: T_p M \rightarrow T_{f(p)} N$ is a linear isometry.

Definition 2.21. A local isometry $f: (M, g_M) \rightarrow (N, g_N)$ between Riemannian manifolds is called an *isometry* if it is a diffeomorphism.

Proposition 2.22. *Let $f: (M, g_M) \rightarrow (N, g_N)$ be smooth map between Riemannian manifolds. Then the following statements are equivalent.*

- (1) f is a local isometry.
- (2) For each $p \in M$, there are open neighborhoods U of p and V of $f(p)$ such that $f|_U: U \rightarrow V$ is an isometry.

Proof. it is clear (2) implies (1), and the converse follows immediately from inverse function theorem. $\square$

Definition 2.23. The *isometry group* $\text{Iso}(M, g)$ of Riemannian manifold (M, g) is the group consisting of all isometries from (M, g) to itself.

Proposition 2.24 ([Pet06, Proposition 5.6.1]). *Let $f: (M, g_M) \rightarrow (N, g_N)$ be a local isometry.*

- (1) f maps geodesics to geodesics.
- (2) $f \circ \exp_p = \exp_{f(p)} \circ (df)_p$ holds on V_p .
- (3) f is distance decreasing.
- (4) If f is an isometry, then f is distance preserving.

Theorem 2.25 ([MS39]). Let (M, g) be a Riemannian manifold and $G = \text{Iso}(M, g)$. Then

- (1) G is a Lie group with respect to compact-open topology.
- (2) For each $p \in M$, the isotropy group G_p is compact.
- (3) G is compact if M is compact.

2.3 Motivation of connection

Given a vector bundle E on a smooth manifold M , the motivation of connection on E arises from how to take the "derivative" of a section s of E in a given direction. It is quite natural to ask such a question, since when we learn calculus, we learn how to take the derivative of a smooth function $f: M \rightarrow \mathbb{R}^m$, to obtain a 1-form.

If we change our language in a more fancy way, any smooth function $f: M \rightarrow \mathbb{R}^m$ can be thought as a section of the trivial vector bundle $M \times \mathbb{R}^m$ as follows:

$$x \mapsto (x, f(x)),$$

and its derivative df can be thought a section of $T^*M \otimes (M \times \mathbb{R}^m)$. Then taking derivatives can be seen as the following operator:

$$\nabla: C^\infty(M, M \times \mathbb{R}^m) \rightarrow C^\infty(M, T^*M \otimes (M \times \mathbb{R}^m)).$$

This motivates us why we define a connection on a vector bundle as follows:

Definition 2.26. Let E be a vector bundle on a smooth manifold M . A *connection* ∇ on a vector bundle E is a $\mathbb{R}$ -linear operator

$$\nabla: C^\infty(M, E) \rightarrow C^\infty(M, T^*M \otimes E)$$

satisfying the Leibniz rule $\nabla(fs) = df \otimes s + f\nabla s$, where $f \in C^\infty(M)$ and $s \in C^\infty(M, E)$.

Remark 2.27 (local form). Let $\{e_\alpha\}$ be a local frame of E . For any local section $s = s^\alpha e_\alpha$ of E , the Leibniz rule implies

$$\nabla(s^\alpha e_\alpha) = ds^\alpha \otimes e_\alpha + s^\alpha \nabla e_\alpha.$$

Thus, the connection ∇ is determined by $\nabla e_\alpha = \omega_\alpha^\beta \otimes e_\beta$, where ω_α^β are 1-forms. For simplicity, we always omit the tensor notion $\otimes$ if there is no ambiguity.

The collection of 1-forms $\{\omega_\alpha^\beta\}$ forms a matrix ω , with the lower index is the row index and the upper index is the column index.

Remark 2.28. Let $\{e_\alpha\}$ be a local frame of vector bundle E and ω be the matrix of connection 1-forms. If we write $\omega_\alpha^\beta = \Gamma_{i\alpha}^\beta dx^i$, then $\Gamma_{i\alpha}^\beta$ is called the *Christoffel symbol*.

Let X be a vector field which is locally written as $X = X^i \frac{\partial}{\partial x^i}$ and s be a section of E which is locally written as $s^\alpha e_\alpha$. Then

$$\begin{aligned}\nabla_X s &= \nabla_{X^i \frac{\partial}{\partial x^i}} s^\alpha e_\alpha \\ &= X^i \nabla_{\frac{\partial}{\partial x^i}} s^\alpha e_\alpha \\ &= X^i s^\alpha \nabla_{\frac{\partial}{\partial x^i}} e_\alpha + X^i \frac{\partial s^\alpha}{\partial x^i} e_\alpha \\ &= (X^i s^\alpha \Gamma_{i\alpha}^\beta + X(s^\beta)) e_\beta,\end{aligned}$$

which is called the *covariant derivative* of s with respect to X .

2.4 Levi-Civita connection

2.4.1 Compatibility with metric

Now consider a vector bundle E with a metric g , which can be locally written as $g_{\alpha\beta} e^\alpha \otimes e^\beta$. If there is a connection ∇ on E , it is natural to ask it to be compatible with metric.

Definition 2.29. A connection ∇ on a vector bundle E is *compatible with the metric* g if for any two sections s, t of E , we have

$$dg(s, t) = g(\nabla s, t) + g(s, \nabla t).$$

Remark 2.30 (local form). Let $\{e_\alpha\}$ be a local frame of E . A direct computation shows

$$\begin{aligned}dg_{\alpha\beta} &= dg(e_\alpha, e_\beta) \\ &= g(\nabla e_\alpha, e_\beta) + g(e_\alpha, \nabla e_\beta) \\ &= \omega_\alpha^\gamma g_{\gamma\beta} + g_{\alpha\gamma} \omega_\beta^\gamma\end{aligned}$$

for all α, β , which is equivalent to

$$\frac{\partial}{\partial x^i} g_{\alpha\beta} = \Gamma_{i\alpha}^\gamma g_{\gamma\beta} + \Gamma_{i\beta}^\gamma g_{\alpha\gamma}$$

for all i, α, β . In matrix notation we have

$$dg = \omega g + g \omega^t.$$

2.4.2 Torsion-freeness

Now let's choose our vector bundle E to be the tangent bundle of a Riemannian manifold (M, g) .

Definition 2.31. A connection ∇ of TM is *torsion-free* if

$$\nabla_X Y - \nabla_Y X = [X, Y],$$

where X, Y are vector fields.

Remark 2.32 (local form). If we choose $X = \frac{\partial}{\partial x^i}, Y = \frac{\partial}{\partial x^j}$, then

$$\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} - \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^i} = (\Gamma_{ij}^k - \Gamma_{ji}^k) \frac{\partial}{\partial x^k} = 0,$$

which is equivalent to Γ_{ij}^k is symmetric in i and j .

2.4.3 Levi-Civita connection

There are infinitely many connection on the tangent bundle of a Riemannian manifold, but an interesting thing is that there is a unique torsion-free connection which is compatible with the Riemannian metric.

Note that compatibility with Riemannian metric g implies

$$Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$$

$$Yg(Z, X) = g(\nabla_Y Z, X) + g(Z, \nabla_Y X)$$

$$Zg(X, Y) = g(\nabla_Z X, Y) + g(X, \nabla_Z Y),$$

where X, Y and Z are vector fields. Adding the first two equations, subtracting the third one, and using the torsion-free condition, we will see

$$Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) = g([X, Z], Y) + g([Y, Z], X) + g([X, Y], Z) + 2g(Z, \nabla_Y X),$$

and thus

$$g(Z, \nabla_Y X) = \frac{1}{2}(Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) - g([X, Z], Y) - g([Y, Z], X) - g([X, Y], Z)),$$

which implies $\nabla_X Y$ is uniquely determined. The above formula is also called the *Koszul formula*.

Remark 2.33 (local form). Firstly, compatibility implies

$$\frac{\partial g_{ij}}{\partial x^k} = \Gamma_{ki}^\ell g_{\ell j} + \Gamma_{kj}^\ell g_{i\ell}.$$

By permuting i, j, k we obtain the following two equations:

$$\frac{\partial g_{jk}}{\partial x^i} = \Gamma_{ij}^\ell g_{\ell k} + \Gamma_{ik}^\ell g_{j\ell}$$

$$\frac{\partial g_{ki}}{\partial x^j} = \Gamma_{jk}^\ell g_{\ell i} + \Gamma_{ji}^\ell g_{k\ell}.$$

By the symmetry of Γ_{ij}^ℓ in i, j and the symmetry of g_{ij} , we have

$$2\Gamma_{ij}^\ell g_{\ell k} = \frac{\partial g_{kj}}{\partial x^i} + \frac{\partial g_{ik}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^k}.$$

If we use (g^{ij}) to denote the inverse matrix of (g_{ij}) , then we have

$$\Gamma_{ij}^\ell = \frac{1}{2} g^{k\ell} \left(\frac{\partial g_{kj}}{\partial x^i} + \frac{\partial g_{ik}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^k} \right),$$

which implies the Christoffel symbol is completely determined by the Riemannian metric and its first order partial derivatives.

3 Tensor computation

3.1 Induced connection on tensor

Let M be a smooth manifold and ∇ be a connection on the tangent bundle TM . In this section, we will introduce how to construct a connection on a (r, s) -tensor.

Firstly, let's see how to induce a connection on T^*M . Since there is a natural non-degenerate pairing $(\cdot, \cdot)$ between TM and T^*M , the *induced connection* on T^*M should respect this natural pairing. To be precise, for any $X \in C^\infty(M, TM)$ and $\omega \in C^\infty(M, T^*M)$, the induced connection T^*M is defined by the following equality

$$d(\omega, X) = (\nabla^{T^*M} \omega, X) + (\omega, \nabla X).$$

Suppose $\{\frac{\partial}{\partial x^i}\}$ is a local frame of TM with dual frame $\{dx^i\}$ of T^*M . Then

$$0 = ((\omega^{T^*M})_k^i dx^k, \frac{\partial}{\partial x^j}) + (dx^i, (\omega)_j^k \frac{\partial}{\partial x^k}),$$

that is,

$$\nabla dx^k = -\Gamma_{ij}^k dx^i \otimes dx^j,$$

where Γ_{ij}^k is the Christoffel symbol of ∇^{TM} .

Remark 3.1 (another viewpoint for torsion-freeness). For any section ω of T^*M , there is a natural 2-form obtained from taking the exterior derivative $d\omega$, and note that $\wedge^2 T^*M$ is just the skew-symmetrization of $T^*M \otimes T^*M$, so it is natural to require the skew-symmetrization of $\nabla \omega$ is $d\omega$. If we check this locally, it suffices to require the skew-symmetrization of ∇dx^k is zero, that is,

$$-\Gamma_{ij}^k dx^i \wedge dx^j = 0.$$

In other words, connection ∇ is torsion-free if and only if the skew-symmetrization of $\nabla \omega$ is $d\omega$.

Definition 3.2. Given a (r, s) -tensor T .

(1) The *covariant derivative* ∇T is a $(r+1, s)$ -tensor, which is defined by

$$\begin{aligned}\nabla T(X_1, X_2, \dots, X_{r+1}, \omega_1, \dots, \omega_s) := & XT(X_1, \dots, X_r, \omega_1, \dots, \omega_s) \\ & - \sum_{k=1}^r T(X_1, \dots, \nabla_{X_1} X_k, \dots, X_r, \omega_1, \dots, \omega_s) \\ & - \sum_{\ell=1}^s T(X_1, \dots, X_r, \omega_1, \dots, \nabla_{X_1} \omega_\ell, \dots, \omega_s),\end{aligned}$$

where $X_1, \dots, X_{r+1}$ are vector fields and $\omega_1, \dots, \omega_s$ are 1-forms.

(2) For a vector field X , the *covariant derivative of T with respect to vector field X* , denoted by $\nabla_X T$ is a (r, s) -tensor, defined by

$$\nabla_X T(X_1, \dots, X_r, \omega_1, \dots, \omega_s),$$

where $X_1, \dots, X_r$ are vector fields and $\omega_1, \dots, \omega_s$ are 1-forms.

Remark 3.3 (local form). If we write a (r, s) -tensor T locally as

$$T_{i_1 \dots i_r}^{j_1 \dots j_s} dx^{i_1} \otimes \dots \otimes dx^{i_r} \otimes \frac{\partial}{\partial x^{j_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{j_s}},$$

and $(r+1, s)$ -tensor ∇T locally as

$$\nabla_i T_{i_1 \dots i_r}^{j_1 \dots j_s} dx^i \otimes dx^{i_1} \otimes \dots \otimes dx^{i_r} \otimes \frac{\partial}{\partial x^{j_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{j_s}}.$$

Then by definition we have

$$\nabla_i T_{i_1 \dots i_r}^{j_1 \dots j_s} = \frac{\partial T_{i_1 \dots i_r}^{j_1 \dots j_s}}{\partial x^i} - \sum_{k=1}^r \Gamma_{ii_k}^q T_{i_1 \dots i_{k-1} q i_{k+1} \dots i_r}^{j_1 \dots j_s} + \sum_{\ell=1}^s \Gamma_{iq}^{j_\ell} T_{i_1 \dots i_r}^{j_1 \dots j_{\ell-1} q j_{\ell+1} \dots j_s}.$$

Definition 3.4. For a (r, s) -tensor T , the *second covariant derivative* is defined as $\nabla^2 T := \nabla(\nabla T)$, which is a $(r+2, s)$ -tensor.

Given vector fields X, Y and a (r, s) -tensor T , firstly we consider its covariant derivative with respect to direction Y , it gives a (r, s) -tensor $\nabla_Y T$, and then we consider the covariant derivative of $\nabla_Y T$ with respect to the direction X , it gives a (r, s) -tensor $\nabla_X \nabla_Y T$.

Since both $\nabla_{X,Y}^2 T$ and $\nabla_X \nabla_Y T$ are (r, s) -tensors, a natural question is what is the relation between them? The answer is they're not the same, and the main reason is that the former is linear over smooth functions in Y , while the latter is not. Let's start with an easy example:

Example 3.5. Given $(0, 0)$ -tensor f , that is, a smooth function f .

(1) The covariant derivative ∇f is an $(1, 0)$ -tensor, which is given by

$$\nabla f = \nabla_i f dx^i = \frac{\partial f}{\partial x^i} dx^i.$$

In other words, ∇f coincides with df .

- (2) The second covariant derivative $\nabla^2 f$ is called the *Hessian* of f , and usually it is denoted by $\text{Hess } f$. More explicitly, we have

$$\begin{aligned}\text{Hess } f &= \nabla(\nabla_i f dx^i) \\ &= \frac{\partial \nabla_i f}{\partial x^k} dx^k \otimes dx^i - \nabla_i f \Gamma_{kj}^i dx^k \otimes dx^j \\ &= \left(\frac{\partial^2 f}{\partial x^k \partial x^i} - \Gamma_{ki}^j \frac{\partial f}{\partial x^j} \right) dx^k \otimes dx^i,\end{aligned}$$

$$\text{that is, } \nabla_{k,i}^2 f = \frac{\partial^2 f}{\partial x^k \partial x^i} - \Gamma_{ki}^j \frac{\partial f}{\partial x^j}.$$

This shows that

$$\nabla_{k,i}^2 f = \nabla_k \nabla_i f - \Gamma_{ki}^j \frac{\partial f}{\partial x^j}.$$

The following proposition shows the relation:

Proposition 3.6. *For any (r, s) -tensor T and vector fields X, Y ,*

$$\nabla_{X,Y}^2 T = \nabla_X \nabla_Y T - \nabla_{(\nabla_X Y)} T.$$

Proof. Let $\{\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\}$ be a local frame of TM and $\{dx^1, \dots, dx^n\}$ be the dual frame of T^*M . Suppose

$$\nabla^2 T = \nabla_{k,i}^2 T_{i_1 \dots i_r}^{j_1 \dots j_s} dx^k \otimes dx^i \otimes dx^{i_1} \otimes \dots \otimes dx^{i_r} \otimes \frac{\partial}{\partial x^{j_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{j_s}}$$

and

$$\nabla_k \nabla_i T = \nabla_k \nabla_i T_{i_1 \dots i_r}^{j_1 \dots j_s} dx^{i_1} \otimes \dots \otimes dx^{i_r} \otimes \frac{\partial}{\partial x^{j_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{j_s}}.$$

It suffices to prove

$$\nabla_{k,i}^2 T_{i_1 \dots i_r}^{j_1 \dots j_s} = \nabla_k \nabla_i T_{i_1 \dots i_r}^{j_1 \dots j_s} - \Gamma_{ki}^j \nabla_j T_{i_1 \dots i_r}^{j_1 \dots j_s}.$$

A direct computation shows

$$\begin{aligned}
\nabla_{k,i}^2 T_{i_1 \dots i_r}^{j_1 \dots j_s} &= \nabla^2 T \left(\frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right) \\
&= \nabla_{\frac{\partial}{\partial x^k}} \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right) \\
&= \underbrace{\frac{\partial}{\partial x^k} \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right)}_{\text{part I}} \\
&\quad - \underbrace{\nabla T \left(\nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right)}_{\text{part II}} \\
&\quad - \underbrace{\sum_{\ell=1}^s \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, \nabla_{\frac{\partial}{\partial x^k}} dx^{j_\ell}, \dots, dx^{j_s} \right)}_{\text{part III}} \\
&\quad - \underbrace{\sum_{m=1}^r \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^{i_m}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right)}_{\text{part IV}}.
\end{aligned}$$

Note that Part II is $-\Gamma_{ki}^j \nabla_j T_{i_1 \dots i_r}^{j_1 \dots j_s}$. On the other hand, the covariant derivative $\nabla_i T$ is defined by

$$\nabla_i T = \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right),$$

and thus by definition

$$\begin{aligned}
\nabla_k \nabla_i T_{i_1 \dots i_r}^{j_1 \dots j_s} &= \frac{\partial}{\partial x^k} \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right) \\
&\quad - \sum_{\ell=1}^s \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, \nabla_{\frac{\partial}{\partial x^k}} dx^{j_\ell}, \dots, dx^{j_s} \right) \\
&\quad - \sum_{m=1}^r \nabla T \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^{i_1}}, \dots, \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^{i_m}}, \dots, \frac{\partial}{\partial x^{i_r}}, dx^{j_1}, \dots, dx^{j_s} \right).
\end{aligned}$$

This shows $\nabla_k \nabla_i T_{i_1 \dots i_r}^{j_1 \dots j_s}$ coincides with Part I + part III + part IV. $\square$

Proposition 3.7. *Let g be a Riemannian metric. Then a connection ∇ is compatible with g if and only if g is parallel, that is, $\nabla g = 0$.*

3.2 Type change of tensor

In this section, we will introduce, how to change a tensor of type (r, s) to another tensor of type $(s - k, r + k)$ for all k such that $s - k \geq 0, r + k \geq 0$, by music isomorphism.

Firstly we have to introduce the induced metric on

Let (M, g) be a Riemannian manifold. Firstly, we have to introduce the induce metric on T^*M .

Recall that a Riemannian metric g is an $(2, 0)$ -tensor, locally written as

$$g = g_{ij} dx^i \otimes dx^j.$$

As the inverse of a positive definite symmetric matrix is again a positive definite symmetric matrix, it induces a metric on T^*M , locally given by

$$g^{ij} \frac{\partial}{\partial x^i} \otimes \frac{\partial}{\partial x^j}.$$

For any vector field X , it gives a 1-form by

$$X^\flat: Y \mapsto g(X, Y),$$

where Y is a vector field. Let $\{\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}\}$ be a local frame of TM and $\{dx^1, \dots, dx^n\}$ be the dual frame of T^*M . If we write $(\frac{\partial}{\partial x^i})^\flat = A_j dx^j$, then

$$A_j = g(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}) = g_{ij}.$$

This shows $(\frac{\partial}{\partial x^i})^\flat = g_{ij} dx^j$ of T^*M .

For any 1-form ω , it can be regarded as a section of $TM = T^{**}M$ as follows

$$\omega^\sharp: \beta \mapsto g(\omega, \beta),$$

If we write $(dx^i)^\sharp = B^j \frac{\partial}{\partial x^j}$, then

$$B^j = g(dx^i, dx^j) = g^{ij}.$$

This shows $(dx^i)^\sharp = g^{ij} \frac{\partial}{\partial x^j}$.

In a summary, we have the so-called *music isomorphisms*, which locally look like

$$\begin{aligned} \flat: C^\infty(M, TM) &\rightarrow C^\infty(M, T^*M) & \sharp: C^\infty(M, T^*M) &\rightarrow C^\infty(M, TM) \\ \frac{\partial}{\partial x^i} &\mapsto g_{ij} dx^j, & dx^j &\mapsto g^{ij} \frac{\partial}{\partial x^i}. \end{aligned}$$

Example 3.8. For a smooth function f , ∇f is an $(1, 0)$ -tensor, locally written as

$$\nabla f = \frac{\partial f}{\partial x^i} dx^i.$$

Then we can change it to a tensor of type $(0, 1)$, that is,

$$\nabla f = g^{ij} \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^j}.$$

It is called *gradient* of f . More generally, for a 1-form ω , locally looks like $\omega_i dx^i$. Then we can change it in an $(1,0)$ -tensor, and it locally looks like

$$\omega^\sharp = g^{ij} \omega_i \frac{\partial}{\partial x^j}.$$

An application of type change is, we can use it to define the trace of an arbitrary tensor. Let's see a simple example: For an $(1,1)$ -tensor T , we can define its "trace", since there is a natural isomorphism between $TM \otimes T^*M$ and $\text{End}(TM)$. Thus, we can take its trace in the sense of matrix. To be explicit, if we locally write T as $T = T_j^i \frac{\partial}{\partial x^i} \otimes dx^j$, then trace of T , denoted by $\text{tr}_g T$, is defined as T_i^i .

If T is not of $(1,1)$ -type, then we change it into $(1,1)$ -type and then take trace:

- (1) If $T = T_{ij} dx^i \otimes dx^j$, then $T = g^{ik} T_{ij} dx^j \otimes \frac{\partial}{\partial x^k}$. Thus, $\text{tr}_g T = g^{ij} T_{ij}$.
- (2) If $T = T^{ij} \frac{\partial}{\partial x^i} \otimes \frac{\partial}{\partial x^j}$, then $T = g_{kj} T^{ij} dx^k \otimes \frac{\partial}{\partial x^i}$, that is, $\text{tr}_g T = g_{ij} T^{ij}$.

In general, if a tensor of type (r,s) with $r+s = 2n$, we can change its type into (n,n) and take trace n times to obtain a number. Later we will see we obtain Ricci curvature by taking trace of curvature, and we obtain scalar curvature by taking trace of Ricci curvature.

Remark 3.9. For a smooth function $f: M \rightarrow \mathbb{R}$, $\nabla^2 f$ is an $(2,0)$ -form, locally looks like

$$\nabla_{i,j}^2 f dx^i \otimes dx^j.$$

Its trace looks like

$$\text{tr}_g \nabla^2 f = g^{ij} \nabla_{i,j}^2 f.$$

That's called the *scalar Laplacian* of f , which is denoted by Δf .

Proposition 3.10 (magic formula). *For an $(2,0)$ -tensor T , we have*

$$X(\text{tr}_g T) = g(g, \nabla_X T).$$

Remark 3.11 (local form). The magic formula can be locally written as

$$\nabla_i (g^{jk} T_{jk}) = g^{jk} (\nabla_i T_{jk}),$$

that is, g^{jk} can "pass through" the covariant derivative, and that is why we call it "magic formula".

4 Geodesic and normal coordinate

In this section we always assume (M, g) is a Riemannian manifold equipped with Levi-Civita connection ∇ .

4.1 Geodesic

Definition 4.1. A smooth curve $\gamma: (-\varepsilon, \varepsilon) \rightarrow M$ is called a *geodesic*, if for each local coordinate $\{x^i\}$, it satisfies

$$\frac{d^2\gamma^k}{dt^2} + \frac{d\gamma^i}{dt} \frac{d\gamma^j}{dt} \Gamma_{ij}^k \circ \gamma = 0,$$

where $\gamma^i = x^i \circ \gamma$.

Theorem 4.2. For any $p \in M, v \in T_p M$, there exists $\varepsilon > 0$ and a geodesic $\gamma: (-\varepsilon, \varepsilon) \rightarrow M$ such that

$$\gamma(0) = p,$$

$$\gamma'(0) = v.$$

Moreover, any two such geodesics agree on their common domain.

Proof. It follows from standard results about existence and uniqueness in ODEs' theory. $\square$

Remark 4.3. Note that standard result in ODEs' theory only guarantees the short time existence of geodesic. If we use I to denote the maximal interval such that γ can be defined on it, then in general $I \neq \mathbb{R}$.

Notation 4.4. For $v \in T_p M$, γ_v denotes the unique geodesic such that $\gamma(0) = p$ and $\gamma'(0) = v$.

Lemma 4.5. For each $p \in M, v \in T_p M$ and $c, t \in \mathbb{R}$, one has

$$\gamma_{cv}(t) = \gamma_v(ct),$$

whenever either side is defined.

Proof. It is clear by uniqueness. $\square$

Definition 4.6. For any $p \in M$, V_p is a subspace of $T_p M$ defined by

$$V_p := \{v \in T_p M \mid \gamma_v(1) \text{ is defined}\}.$$

Remark 4.7. From Lemma 4.5, we know that $v \in V_p$ if $|v| < \varepsilon$ for sufficiently small $\varepsilon > 0$.

4.2 Normal coordinate

4.2.1 Exponential map and geodesical ball

Definition 4.8. For $p \in M$, the *exponential map* at point p is the map

$$\begin{aligned}\exp_p: V_p &\rightarrow M \\ v &\mapsto \gamma_v(1).\end{aligned}$$

Theorem 4.9. The exponential map $\exp_p$ is a local diffeomorphism at $0 \in T_p M$, that is, there exists a neighborhood V of $0 \in T_p M$ which is mapped diffeomorphically to a neighborhood U of $p \in M$.

Proof. Note that

$$(\mathrm{d}\exp_p)_0: T_0(T_p M) \rightarrow T_p M$$

and if we identify $T_0(T_p M)$ with $T_p M$ by using $0 + tv \mapsto v$, then $(\mathrm{d}\exp_p)_0$ is a linear map from $T_p M$ to itself. By inverse function theorem, it suffices to check $(\mathrm{d}\exp_p)_0$ is an isomorphism. For all $v \in T_p M$, a direct computation shows

$$\begin{aligned}(\mathrm{d}\exp_p)_0(v) &\stackrel{(1)}{=} \left. \frac{d}{dt} \right|_{t=0} \exp_p(0 + tv) \\ &= \left. \frac{d}{dt} \right|_{t=0} \gamma_{tv}(1) \\ &= \left. \frac{d}{dt} \right|_{t=0} \gamma_v(t) \\ &= \gamma'_v(0) \\ &= v,\end{aligned}$$

where (1) holds from identification $T_0(T_p M) \cong T_p M$. □

Definition 4.10. For $p \in M$, if $B(0, \delta)$ is a ball in $T_p M$ such that $\exp_p$ is a diffeomorphism on $B(0, \delta)$, then $B(p, \delta) := \exp_p(B(0, \delta))$ is called a *geodesic ball* centered at p .

Theorem 4.11. For each $p \in M$, there exists a local coordinate centered at p , called *normal coordinate*, such that we have the following expressions:

- (1) $g_{ij}(0) = \delta_{ij}$;
- (2) $\Gamma_{ij}^k(0) = 0$.

Proof. Let $\{\frac{\partial}{\partial x^1}|_p, \dots, \frac{\partial}{\partial x^n}|_p\}$ be an orthonormal basis of $T_p M$ with respect to inner product $g(p)$, and fix the following linear isomorphism

$$\begin{aligned}\Phi: T_p M &\rightarrow \mathbb{R}^n \\ v^i \frac{\partial}{\partial x^i} \Big|_p &\mapsto (v^1, \dots, v^n).\end{aligned}$$

By Theorem 4.9 there exists a neighborhood U of p which is mapped by $\Phi \circ \exp_p^{-1}$ diffeomorphically onto a neighborhood V of $0 \in \mathbb{R}^n$, so $\Phi \circ \exp_p^{-1}: U \rightarrow V$ gives a local coordinate centered at p . This is called the normal coordinate. Given a local coordinate $\varphi: U \rightarrow V$ centered at $p \in M$, we have the following diagram

$$\begin{array}{ccc} T_p M \times T_p M & \longrightarrow & \mathbb{R} \\ (\mathrm{d}\varphi)_p \times (\mathrm{d}\varphi)_p \downarrow & & \downarrow \\ \mathbb{R}^n \times \mathbb{R}^n & \longrightarrow & \mathbb{R}, \end{array}$$

and we can use this diagram to express Riemannian metric and Christoffel symbols as a function defined on $V \subseteq \mathbb{R}^n$.

For (1). Let e_i denotes $(0, \dots, \underbrace{1}_{i\text{-th}}, \dots, 0) \in \mathbb{R}^n$. Then

$$\begin{aligned} g_{ij}(0) &:= \left\langle (\mathrm{d}\exp_p \circ \Phi^{-1})_0 e_i, (\mathrm{d}\exp_p \circ \Phi^{-1})_0 e_j \right\rangle_p \\ &\stackrel{(1)}{=} \left\langle (\mathrm{d}\exp_p)_0 \frac{\partial}{\partial x^i} \Big|_p, (\mathrm{d}\exp_p)_0 \frac{\partial}{\partial x^j} \Big|_p \right\rangle_p \\ &\stackrel{(2)}{=} \left\langle \frac{\partial}{\partial x^i} \Big|_p, \frac{\partial}{\partial x^j} \Big|_p \right\rangle_p \\ &\stackrel{(3)}{=} \delta_{ij}, \end{aligned}$$

where

1. (1) holds from Φ is a linear map, thus $\mathrm{d}\Phi^{-1} = \Phi^{-1}$.
2. (2) holds from Theorem 4.9.
3. (3) holds from $\{\frac{\partial}{\partial x^i} \Big|_p\}$ is an orthonormal basis with respect to $g(p)$.

For (2). For $v = (v^1, \dots, v^n) \in \mathbb{R}^n$, by definition $\gamma(t) = \exp_p(t\Phi^{-1}(v))$ is the geodesic with $\gamma(0) = p$ and $\gamma'(0) = \Phi^{-1}(v)$. In normal coordinate (x^i, U, p) , one has $\Phi \circ \exp_p^{-1}(\gamma(t)) = (tv^1, \dots, tv^n)$. Since the Riemannian metric on V is given by the one on U , that is, $\Phi \circ \exp_p^{-1}$ is an isometry, we have $\Phi \circ \exp_p^{-1}(\gamma(t))$ is a geodesic, and the geodesic equation is given by

$$\Gamma_{ij}^k(tv)v^i v^j = 0.$$

By evaluating this expression at $t = 0$ one has $\Gamma_{ij}^k(0)v^i v^j = 0$ for arbitrary index k , and take $v = e_i + e_j$ to conclude $\Gamma_{ij}^k(0) = 0$ for all i, j, k . $\square$

Remark 4.12. Tensor computation is one of the hallmarks of Riemannian geometry, but the normal coordinate can help you to simplify it. The idea is, if we want to prove an identity of tensors, it suffices to choose a coordinate and check it pointwisely. Thus, the normal coordinate is a good

choice, since by Theorem 4.11, one has the Riemannian metric is standard metric, and Christoffel symbols vanish.

Corollary 4.13. *In the normal coordinate, we have the following identities*

$$\frac{\partial \Gamma_{ij}^k}{\partial x^\ell}(0) + \frac{\partial \Gamma_{\ell i}^k}{\partial x^j}(0) + \frac{\partial \Gamma_{j\ell}^k}{\partial x^i}(0) = 0.$$

Proof. As shown in the proof of Theorem 4.11, in normal coordinate we have

$$0 = \Gamma_{ij}^k(tv)v^i v^j.$$

Then by taking differential with respect to x^ℓ it gives

$$0 = \frac{\partial \Gamma_{ij}^k}{\partial x^\ell}(tx)tx^i x^j,$$

which implies

$$\sum_{\sigma \in S_3} \frac{\partial \Gamma_{\sigma(i)\sigma(j)}^k}{\partial x^{\sigma(\ell)}}(0) = 0.$$

Then the desired result holds from the symmetry of Christoffel symbol in term i, j . □

Corollary 4.14. *In normal coordinate (x^i, U, p) , the Taylor expansion of g_{ij} around zero is*

$$g_{ij}(x) = \delta_{ij} + O(|x|^2).$$

Proof. It follows from

$$\frac{\partial g_{ij}}{\partial x^k} = \Gamma_{ki}^\ell(0)g_{\ell j}(0) + \Gamma_{kj}^\ell(0)g_{i\ell}(0) = 0.$$

□

4.3 Hopf-Rinow's Theorem

In this section we will figure out when the exponential map is defined on the whole space $T_p M$.

Definition 4.15. A Riemannian manifold M is *geodesically complete* if for all $p \in M$, the exponential map $\exp_p$ is defined on the whole $T_p M$.

At this stage it is convenient to introduce a distance function on a Riemannian manifold M which is not necessarily geodesic complete as follows: For $p, q \in M$, consider all the piecewise smooth curves joining p and q . Since M is connected, such curves always exist (cover a continuous curve joining p and q by a finite number of coordinates neighborhood and replace each piece contained in a coordinate neighborhood by a smooth one).

Definition 4.16. Let (M, g) be a Riemannian manifold, $p, q \in M$, the *distance* between p and q is defined by the infimum of the lengths of all piecewise smooth curves joining p and q , denoted by $\text{dist}(p, q)$.

Proposition 4.17 ([dC92, Proposition 2.6]). *The topology induced by distance function on M coincides with the original topology on M .*

Theorem 4.18 (Hopf-Rinow). *Let (M, g) be a Riemannian manifold and $p \in M$. The following statements are equivalent.*

- (1) *M is geodesically complete.*
- (2) *The closed and bounded sets of M are compact.*
- (3) *M is complete as a topological space.*

In addition, any of statements above implies that for any $p, q \in M$, there exists a geodesic joining p and q with length $\text{dist}(p, q)$.

Proof. See [dC92, Theorem 2.8]. □

Remark 4.19. Note that (2) is equivalent to (3) is a basic fact in general topology.

Definition 4.20. A Riemannian manifold is *complete*, if it is geodesically complete.

Corollary 4.21. *If M is compact, then it is complete.*

5 Riemannian Curvature

5.1 Curvature form and curvature tensor

5.1.1 Curvature form

Let (M, g) be a Riemannian manifold with connection ∇ of a vector bundle E over M . Now we're going to extend the connection to something called the exterior derivative defined on sections of vector bundle valued k -forms as follows

$$\begin{aligned} d^\nabla: C^\infty(M, \Omega_M^k \otimes E) &\rightarrow C^\infty(M, \Omega_M^{k+1} \otimes E) \\ \omega \otimes e &\mapsto d\omega \otimes e + (-1)^k \omega \wedge \nabla e. \end{aligned}$$

Suppose $\{e_\alpha\}$ is a local frame of E . Then

$$\begin{aligned}
(d^\nabla)^2(s^\alpha e_\alpha) &= d^\nabla(ds^\alpha \otimes e_\alpha + s^\alpha \omega_\alpha^\beta \otimes e_\beta) \\
&= -ds^\alpha \wedge \omega_\alpha^\beta \otimes e_\beta + d(s^\alpha \omega_\alpha^\beta) \otimes e_\beta - s^\alpha \omega_\alpha^\beta \wedge \omega_\beta^\gamma \otimes e_\gamma \\
&= s^\alpha (d\omega_\alpha^\beta - \omega_\alpha^\gamma \wedge \omega_\gamma^\beta) \otimes e_\beta. \\
(d^\nabla)^2(e_\alpha) &= d^\nabla(\omega_\alpha^\beta \otimes e_\beta) \\
&= d\omega_\alpha^\beta \otimes e_\beta - \omega_\alpha^\beta \wedge \nabla e_\beta \\
&= d\omega_\alpha^\beta \otimes e_\beta - \omega_\alpha^\beta \wedge \omega_\beta^\gamma \otimes e_\gamma \\
&= (d\omega_\alpha^\beta - \omega_\alpha^\gamma \wedge \omega_\gamma^\beta) \otimes e_\beta.
\end{aligned}$$

This shows $(d^\nabla)^2$ is a global section of $\Omega_M^2 \otimes \text{End } E$, that is, it is compatible with change of basis. Indeed, for two local frames $e, \tilde{e}$ such that $\tilde{e} = ge$, we have

$$\begin{aligned}
g(d^\nabla)^2 e &= (d^\nabla)^2 ge \\
&= (d^\nabla)^2 \tilde{e} \\
&= (d\tilde{\omega} - \tilde{\omega} \wedge \tilde{\omega}) \tilde{e} \\
&= (d\tilde{\omega} - \tilde{\omega} \wedge \tilde{\omega}) ge,
\end{aligned}$$

which implies

$$g^{-1}(d\tilde{\omega} - \tilde{\omega} \wedge \tilde{\omega})g = d\omega - \omega \wedge \omega.$$

In general case, for $s \in C^\infty(M, \Omega_M^k \otimes E)$, locally written as $s = s^\alpha e_\alpha$, where s^α is a k -form, a direct computation also shows

$$(d^\nabla)^2(s^\alpha e_\alpha) = s^\alpha \wedge (d^\nabla)^2(e_\alpha).$$

Definition 5.1. Let E be a vector bundle over M equipped with connection ∇ . Then there exists a section $\Omega \in C^\infty(M, \Omega_M^2 \otimes \text{End } E)$, called *curvature form*, such that

$$(d^\nabla)^2 s = \Omega \wedge s$$

for all $s \in C^\infty(X, \Omega_M^k \otimes E)$.

Remark 5.2 (local form). Suppose $\{e_\alpha\}$ is a local frame of E and write $\Omega = \Omega_\alpha^\beta e_\beta \otimes e^\alpha$, where

$$\Omega_\alpha^\beta = \Omega_{ij\alpha}^\beta dx^i \wedge dx^j.$$

Then $\Omega = d\omega - \omega \wedge \omega$ shows

$$\begin{aligned}
\Omega_{ij\alpha}^\beta dx^i \wedge dx^j &= \Omega_\alpha^\beta \\
&= d\omega_\alpha^\beta - \omega_\alpha^\gamma \wedge \omega_\gamma^\beta \\
&= d(\Gamma_{i\alpha}^\beta dx^i) - (\Gamma_{i\alpha}^\gamma dx^i) \wedge (\Gamma_{j\gamma}^\beta dx^j) \\
&= (-\partial_j \Gamma_{i\alpha}^\beta - \Gamma_{i\alpha}^\gamma \Gamma_{j\gamma}^\beta) dx^i \wedge dx^j.
\end{aligned}$$

In other words, we have $\Omega_{ij\alpha}^\beta = -(\partial_j \Gamma_{i\alpha}^\beta + \Gamma_{i\alpha}^\gamma \Gamma_{j\gamma}^\beta)$.

Remark 5.3. In physicists' language, a connection is a "field", the curvature is the "strength" of the field, and choosing a local frame is called "fixing the gauge". The reason for these names comes from H. Weyl's work on rewriting Maxwell's equations.

5.1.2 Curvature tensor

Another standard way to define curvature is to define it as follows:

$$R: C^\infty(M, TM) \times C^\infty(M, TM) \times C^\infty(M, E) \rightarrow C^\infty(M, E)$$

$$(X, Y, s) \mapsto R(X, Y)s := \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X, Y]} s.$$

It is easy to check R defines a section of $\bigotimes^2 T^*M \otimes \text{End } E$

Remark 5.4 (local form). Let $\{e_\alpha\}$ be a local frame of E with dual frame $\{e^\alpha\}$. Then locally

$$R = R_{ij\alpha}^\beta dx^i \otimes dx^j \otimes e^\alpha \otimes e_\beta.$$

A direct computation shows that

$$\nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^j}} e_\alpha = \nabla_{\frac{\partial}{\partial x^i}} (\Gamma_{j\alpha}^\beta e_\beta) = \partial_i \Gamma_{j\alpha}^\beta e_\beta + \Gamma_{j\alpha}^\beta \Gamma_{i\beta}^\gamma e_\gamma = (\partial_i \Gamma_{j\alpha}^\beta + \Gamma_{j\alpha}^\gamma \Gamma_{i\gamma}^\beta) e_\beta.$$

Thus

$$R_\alpha^\beta = (\partial_i \Gamma_{j\alpha}^\beta - \partial_j \Gamma_{i\alpha}^\beta + \Gamma_{j\alpha}^\gamma \Gamma_{i\gamma}^\beta - \Gamma_{i\alpha}^\gamma \Gamma_{j\gamma}^\beta) dx^i \otimes dx^j.$$

In particular, when the curvature form is regarded as a tensor, it coincides with curvature tensor since

$$\begin{aligned} \Omega_{ij\alpha}^\beta dx^i \wedge dx^j &= -(\partial_j \Gamma_{i\alpha}^\beta + \Gamma_{i\alpha}^\gamma \Gamma_{j\gamma}^\beta)(dx^i \otimes dx^j - dx^j \otimes dx^i) \\ &= (\partial_i \Gamma_{j\alpha}^\beta + \Gamma_{j\alpha}^\gamma \Gamma_{i\gamma}^\beta - \partial_j \Gamma_{i\alpha}^\beta - \Gamma_{i\alpha}^\gamma \Gamma_{j\gamma}^\beta) dx^i \otimes dx^j \\ &= R_{ij\alpha}^\beta dx^i \otimes dx^j. \end{aligned}$$

5.1.3 Riemannian curvature

Let (M, g) be a Riemannian manifold and ∇ be the Levi-Civita connection on TM . Then the curvature R is a $(3, 1)$ -tensor, which is locally written as

$$R_{ijk}^r dx^i \otimes dx^j \otimes dx^k \otimes \frac{\partial}{\partial x^r}.$$

Now let's give an explicit expression of $R_{ijkl} = g_{lr} R_{ijk}^r$. By definition we have

$$\begin{aligned}
R_{ijkl} &= R\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell}\right) \\
&= \left\langle \nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^k} - \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell} \right\rangle \\
&= \partial_i \left\langle \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell} \right\rangle - \left\langle \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^k}, \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^\ell} \right\rangle \\
&\quad - \left(\partial_j \left\langle \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell} \right\rangle - \left\langle \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k}, \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^\ell} \right\rangle \right) \\
&= \underbrace{\partial_i \left\langle \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell} \right\rangle - \partial_j \left\langle \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell} \right\rangle}_{\text{part I}} \\
&\quad + \underbrace{\left\langle \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^k}, \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^\ell} \right\rangle - \left\langle \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^k}, \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^\ell} \right\rangle}_{\text{part II}}.
\end{aligned}$$

For part II, we have

$$g_{rs}(\Gamma_{ik}^r \Gamma_{j\ell}^s - \Gamma_{jk}^r \Gamma_{i\ell}^s).$$

For part I, note that

$$\begin{aligned}
\partial_i(\Gamma_{jk}^r g_{r\ell}) &= \partial_i \left(\frac{1}{2} g^{rs} (\partial_j g_{ks} + \partial_k g_{js} - \partial_s g_{jk}) g_{r\ell} \right) \\
&= \partial_i \left(\frac{1}{2} \delta_\ell^s (\partial_j g_{ks} + \partial_k g_{js} - \partial_s g_{jk}) \right) \\
&= \frac{1}{2} \partial_i (\partial_j g_{k\ell} + \partial_k g_{j\ell} - \partial_\ell g_{jk}).
\end{aligned}$$

Thus we have part I is

$$\partial_i(\Gamma_{jk}^r g_{r\ell}) - \partial_j(\Gamma_{ik}^r g_{r\ell}) = \frac{1}{2} (\partial_i \partial_k g_{j\ell} + \partial_j \partial_\ell g_{ik} - \partial_i \partial_\ell g_{jk} - \partial_j \partial_k g_{i\ell}).$$

So we have an explicit expression for R_{ijkl}

$$R_{ijkl} = \frac{1}{2} (\partial_i \partial_k g_{j\ell} + \partial_j \partial_\ell g_{ik} - \partial_i \partial_\ell g_{jk} - \partial_j \partial_k g_{i\ell}) + g_{rs}(\Gamma_{ik}^r \Gamma_{j\ell}^s - \Gamma_{jk}^r \Gamma_{i\ell}^s).$$

From this expression, we can see in general curvature depends on second order partial derivatives of metric. Furthermore, there are some skew symmetries and symmetries of R_{ijkl} .

1. $R_{ijkl} = -R_{jilk}$;
2. $R_{ijkl} = -R_{ijlk}$;
3. $R_{ijkl} = R_{klij}$.

Proposition 5.5. *In normal coordinate, we have*

$$g_{ij} = \delta_{ij} - \frac{1}{3}R_{ik\ell j}(0)x^k x^\ell + O(|x|^3).$$

Proof. Equivalently, it suffices to prove that

$$\frac{\partial^2 g_{ij}}{\partial x^\ell \partial x^k}(0) = -R_{ik\ell j}(0).$$

Recall that the compatibility of connection and Riemannian metric gives

$$\frac{\partial g_{ij}}{\partial x^k} = \Gamma_{ki}^m g_{mj} + \Gamma_{kj}^m g_{mi}.$$

Then by taking partial differential with respect to x^ℓ , evaluating at $x = 0$, it gives

$$\frac{\partial^2 g_{ij}}{\partial x^\ell \partial x^k}(0) = \frac{\partial \Gamma_{ki}^m}{\partial x^\ell}(0)g_{mj}(0) + \frac{\partial \Gamma_{kj}^m}{\partial x^\ell}(0)g_{mi}(0),$$

since the Christoffel symbols vanish at $x = 0$.

On the other hand, we have

$$\begin{aligned} R_{ijk\ell}(0) &= \left(\frac{\partial \Gamma_{jk}^m}{\partial x^i}(0) - \frac{\partial \Gamma_{ik}^m}{\partial x^j}(0) \right) g_{m\ell}(0) \\ &\stackrel{(1)}{=} - \left(\frac{\partial \Gamma_{ij}^m}{\partial x^k}(0) + \frac{\partial \Gamma_{ki}^m}{\partial x^j}(0) + \frac{\partial \Gamma_{ik}^m}{\partial x^j}(0) \right) g_{m\ell}(0) \\ &= - \left(\frac{\partial \Gamma_{ij}^m}{\partial x^k}(0) + 2 \frac{\partial \Gamma_{ki}^m}{\partial x^j}(0) \right) g_{m\ell}(0), \end{aligned}$$

where (1) holds from Lemma 4.13. Thus we have

$$\begin{aligned} 2R_{ikj\ell}(0)x^k x^\ell &= - \left(R_{ik\ell j}(0) + R_{j\ell ki}(0) \right) \\ &= \left(\frac{\partial \Gamma_{ik}^m}{\partial x^\ell}(0) + 2 \frac{\partial \Gamma_{i\ell}^m}{\partial x^k}(0) \right) g_{mj}(0)x^k x^\ell + \left(\frac{\partial \Gamma_{j\ell}^m}{\partial x^k}(0) + 2 \frac{\partial \Gamma_{jk}^m}{\partial x^\ell}(0) \right) g_{mi}(0) \\ &= \left(\frac{\partial \Gamma_{ik}^m}{\partial x^\ell}(0)g_{mj}(0) + \frac{\partial \Gamma_{jk}^m}{\partial x^\ell}(0)g_{mi}(0) + \frac{\partial \Gamma_{i\ell}^m}{\partial x^k}(0)g_{mj}(0) + \frac{\partial \Gamma_{j\ell}^m}{\partial x^k}(0)g_{mi}(0) + \frac{\partial \Gamma_{i\ell}^m}{\partial x^k}(0)g_{mj}(0) + \frac{\partial \Gamma_{jk}^m}{\partial x^\ell}(0)g_{mi}(0) \right) \\ &= \left(\frac{\partial \Gamma_{ik}^m}{\partial x^\ell}(0)g_{mj}(0) + \frac{\partial \Gamma_{jk}^m}{\partial x^\ell}(0)g_{mi}(0) + \frac{\partial \Gamma_{i\ell}^m}{\partial x^k}(0)g_{mj}(0) + \frac{\partial \Gamma_{j\ell}^m}{\partial x^k}(0)g_{mi}(0) + \frac{\partial \Gamma_{i\ell}^m}{\partial x^k}(0)g_{mj}(0) + \frac{\partial \Gamma_{jk}^m}{\partial x^\ell}(0)g_{mi}(0) \right) \\ &= 3 \frac{\partial^2 g_{ij}}{\partial x^k \partial x^\ell}(0). \end{aligned}$$

This completes the proof. □

Corollary 5.6. *In Riemannian normal coordinate we have*

1. $g^{ij} = \delta_{ij} + \frac{1}{3}R_{ik\ell j}(0)x^k x^\ell + O(|x|^3)$.
2. $\det(g_{ij}) = 1 - \frac{1}{3}R_{k\ell}x^k x^\ell + O(|x|^3)$.
3. $\sqrt{\det(g_{ij})} = 1 - \frac{1}{6}R_{k\ell}x^k x^\ell + O(|x|^3)$.

Theorem 5.7 (Ricci identity). *Let (M, g) be a Riemannian manifold and T be a (r, s) -tensor, locally written as $T_{i_1 \dots i_r}^{j_1 \dots j_s} dx^{i_1} \otimes \dots \otimes dx^{i_r} \otimes \frac{\partial}{\partial x^{j_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{j_s}}$. Then*

$$\nabla_{k,i}^2 T_{i_1 \dots i_r}^{j_1 \dots j_s} - \nabla_{i,k}^2 T_{i_1 \dots i_r}^{j_1 \dots j_s} = \sum_{\ell=1}^s R_{kiq}^{j_\ell} T_{i_1 \dots i_r}^{j_1 \dots j_{\ell-1} q j_{\ell+1} \dots j_s} - \sum_{m=1}^r R_{kii_m}^q T_{i_1 \dots i_{m-1} q i_{m+1} \dots i_r}^{j_1 \dots j_s}.$$

5.2 Bianchi identities

5.2.1 First Bianchi

Proposition 5.8 (first Bianchi identity). *Let (M, g) be a Riemannian manifold and R be the curvature tensor of Levi-Civita connection. Let X, Y, Z, W be vector fields on M . Then*

$$R(X, Y, Z, W) + R(Y, Z, X, W) + R(Z, X, Y, W) = 0.$$

Proof. It suffices to prove this identity locally. Suppose $X = \frac{\partial}{\partial x^i}, Y = \frac{\partial}{\partial x^j}, Z = \frac{\partial}{\partial x^k}, W = \frac{\partial}{\partial x^\ell}$. Then locally the first Bianchi identity is stated as

$$R_{ijkl} + R_{jkil} + R_{kijl} = 0.$$

In other words, it suffices to prove

$$R_{ijk}^r + R_{jki}^r + R_{kij}^r = 0.$$

Since we have

$$R_{ijk}^r = \underbrace{\partial_i \Gamma_{jk}^r - \partial_j \Gamma_{ik}^r}_{\text{part I}} + \underbrace{\Gamma_{jk}^s \Gamma_{is}^r - \Gamma_{ik}^s \Gamma_{js}^r}_{\text{part II}}.$$

(1) For the first part, if we permuting i, j, k , we have

$$\partial_i \Gamma_{jk}^r - \partial_j \Gamma_{ik}^r + \partial_j \Gamma_{ki}^r - \partial_k \Gamma_{ji}^r + \partial_k \Gamma_{ij}^r - \partial_i \Gamma_{kj}^r = 0,$$

since $\Gamma_{ij}^r = \Gamma_{ji}^r$ by torsion-free.

(2) For the second part, if we permuting i, j, k , we have

$$\Gamma_{jk}^s \Gamma_{is}^r - \Gamma_{ik}^s \Gamma_{js}^r + \Gamma_{ki}^s \Gamma_{js}^r - \Gamma_{ji}^s \Gamma_{ks}^r + \Gamma_{ij}^s \Gamma_{ks}^r - \Gamma_{kj}^s \Gamma_{is}^r = 0$$

by the same reason.

This completes the proof of first Bianchi identity. □

Remark 5.9. If we consider connection on arbitrary vector bundle E , there is no first Bianchi identity, since e_α is just a section of E , not a section of TM , and thus $R(e_\alpha, \cdot)$ or $R(\cdot, e_\alpha)$ is nonsense.

5.2.2 Second Bianchi

Proposition 5.10 (second Bianchi identity). *Let (M, g) be a Riemannian manifold and R be the curvature tensor of Levi-Civita connection. Let X, Y, Z, W be vector fields on M . Then*

$$\nabla_X R(Y, Z, W, T) + \nabla_Y R(Z, X, W, T) + \nabla_Z R(X, Y, W, T) = 0,$$

where $X, Y, Z, W, T \in C^\infty(M, TM)$.

The first approach. Without lose of generality, we may choose the normal coordinate $\{x^i\}$. Let $X = \frac{\partial}{\partial x^i}, Y = \frac{\partial}{\partial x^j}, Z = \frac{\partial}{\partial x^k}, W = \frac{\partial}{\partial x^\ell}, T = \frac{\partial}{\partial x^m}$. Then

$$\begin{aligned} \nabla_X R(Y, Z, W, T) &\stackrel{(1)}{=} \nabla_{\frac{\partial}{\partial x^i}} g \left(\nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^\ell} - \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^\ell}, \frac{\partial}{\partial x^m} \right) \\ &\stackrel{(2)}{=} g \left(\nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^\ell} - \nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^\ell}, \frac{\partial}{\partial x^m} \right), \end{aligned}$$

where

1. (1) holds since $[\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}] = 0$;
2. (2) holds from the compatibility of connection and metric, and Christoffel symbols vanish at p in the normal coordinate.

By permuting the index i, j, k , we have

$$\begin{aligned} &\nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^\ell} - \nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^\ell} \\ &+ \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^\ell} - \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^\ell} \\ &+ \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^i}} \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^\ell} - \nabla_{\frac{\partial}{\partial x^k}} \nabla_{\frac{\partial}{\partial x^j}} \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^\ell} \\ &= R \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right) \nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^\ell} + R \left(\frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k} \right) \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^\ell} + R \left(\frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^i} \right) \nabla_{\frac{\partial}{\partial x^j}} \frac{\partial}{\partial x^\ell} \\ &= 0, \end{aligned}$$

where the last equality holds from $\nabla_{\frac{\partial}{\partial x^k}} \frac{\partial}{\partial x^\ell} = 0$ under the normal coordinate. This completes proof of the second Bianchi identity. $\square$

The second approach. Let Ω be the curvature form of ∇ . Then we claim that

$$d\Omega + \Omega \wedge \omega - \omega \wedge \Omega = 0.$$

Indeed,

$$\begin{aligned} d\Omega + \Omega \wedge \omega - \omega \wedge \Omega &= d(d\omega - \omega \wedge \omega) + (d\omega - \omega \wedge \omega) \wedge \omega - \omega \wedge (d\omega - \omega \wedge \omega) \\ &= d^2\omega - d\omega \wedge \omega + \omega \wedge d\omega + d\omega \wedge \omega - \omega \wedge \omega \wedge \omega - \omega \wedge d\omega + \omega \wedge \omega \wedge \omega \\ &= 0. \end{aligned}$$

If we write this in the local form, it gives

$$(\partial_k \Omega_{ij\alpha}^\beta + \Omega_{ij\alpha}^\gamma \Gamma_{k\gamma}^\beta - \Gamma_{k\alpha}^\gamma \Omega_{ij\gamma}^\beta) dx^k \wedge dx^i \wedge dx^j = 0.$$

By using the relations between curvature tensor and curvature form, this also completes the proof. $\square$

5.3 Other curvatures

5.3.1 Sectional curvature

In this section we're going to introduce sectional curvature, which is used to characterize a two dimensional subspace of tangent space.

Fix $p \in M$ and let x, y are two linearly independent tangent vectors in $T_p M$, then sectional curvature for these two vectors are defined as

$$K_p(x, y) = \frac{R_p(x, y, y, x)}{g_p(x, x)g_p(y, y) - g_p(x, y)^2}.$$

In order to show it is a invariant defined for a two dimensional subspace, we need to check if $\text{span}_{\mathbb{R}}\{x, y\} = \text{span}_{\mathbb{R}}\{z, w\}$, then

$$K_p(x, y) = K_p(z, w).$$

Indeed, if we write

$$\begin{cases} z = ax + by \\ w = cx + dy. \end{cases}$$

Then by symmetry and skew symmetry properties of R we have

$$\begin{aligned} R_p(z, w, w, z) &= R_p(ax + by, cx + dy, cx + dy, ax + by) \\ &= R_p(ax, dy, dy, ax) + R_p(ax, dy, cx, by) + R_p(by, cx, dy, ax) + R_p(by, cx, cx, by) \\ &= a^2 d^2 R_p(x, y, y, x) - abcd R_p(x, y, y, x) - abcd R_p(x, y, y, x) + b^2 c^2 R_p(x, y, y, x) \\ &= (ad - bc)^2 R_p(x, y, y, x). \end{aligned}$$

And by the same computations we have

$$g_p(z, z)g_p(w, w) - g_p(z, w)^2 = (ad - bc)^2 \{g_p(x, x)g_p(y, y) - g_p(x, y)^2\}$$

Thus

$$K_p(x, y) = K_p(z, w).$$

Definition 5.11. The *sectional curvature* $K_p(\sigma)$ for two dimensional subspace $\sigma \subseteq T_p M$ is defined as

$$K_p(\sigma) := K_p(x, y),$$

where $\{x, y\}$ is a basis of σ .

Definition 5.12. A Riemannian manifold (M, g) is called *isotropic*, if for each point $p \in M$, the sectional curvature $K_p(\sigma)$ is independent of σ .

Definition 5.13. A Riemannian manifold (M, g) has *constant sectional curvature*, if $K_p(\sigma)$ is constant for arbitrary $\sigma \subset T_p M, p \in M$.

Remark 5.14. By definition, if a Riemannian manifold has constant sectional curvature, then it must be isotropic; Conversely, if a Riemannian manifold is isotropic and its dimension is ≥ 3 , then it has constant sectional curvature, see Corollary 5.33

Lemma 5.15.

$$-6R(X, Y, Z, W) = \frac{\partial^2}{\partial s \partial t} \Big|_{s=t=0} \{R(X + sZ, Y + tW, Y + tW, X + sZ) - R(X + sW, Y + tZ, Y + tZ, X + sW)\},$$

where X, Y, Z, W are vector fields.

Proof. It suffices to compute coefficients of st of $R(X + sZ, Y + tW, Y + tW, X + sZ)$ and exchange Z with W to obtain coefficients of st of $R(X + sW, Y + tZ, Y + tZ, X + sW)$.

it is easy to see coefficients of st of $R(X + sZ, Y + tW, Y + tW, X + sZ)$ is

$$R(Z, W, Y, X) + R(Z, Y, W, X) + R(X, W, Y, Z) + R(X, Y, W, Z),$$

so coefficients of st of $R(X + sW, Y + tZ, Y + tZ, X + sW)$ is

$$R(W, Z, Y, X) + R(W, Y, Z, X) + R(X, Z, Y, W) + R(X, Y, Z, W).$$

Thus the right hand of our desired identity is

$$-4R(X, Y, Z, W) - (R(Y, Z, W, X) + R(W, Y, Z, X)) - (R(W, X, Y, Z) + R(W, Y, Z, X)).$$

By first Bianchi identity we have

$$\begin{aligned} R(Y, Z, W, X) + R(W, Y, Z, X) &= R(Y, Z, W, X) + R(Z, X, W, Y) \\ &= R(X, Y, Z, W) \\ R(W, X, Y, Z) + R(W, Y, Z, X) &= R(Y, Z, W, X) + R(Z, X, W, Y) \\ &= R(X, Y, Z, W). \end{aligned}$$

This completes the proof. □

Notation 5.16. For convenience, we use $R_0(X, Y, Z, W)$ to denote

$$R_0(X, Y, Z, W) = g(X, W)g(Y, Z) - g(X, Z)g(Y, W),$$

where X, Y, Z, W are vector fields. Then we can write sectional curvature as

$$K_p(\sigma) = \frac{R(x, y, y, x)}{R_0(x, y, y, x)}.$$

where $\sigma \subset T_p M$ is spanned by x, y .

Proposition 5.17. *A Riemannian manifold has constant sectional curvature K_p at point $p \in M$ if and only if the Riemannian curvature tensor $R_p = K_p R_{0,p}$, where K_p is a constant (depending on p).*

Proof. If $R_p = K_p R_{0,p}$, then for a arbitrary x, y , we have

$$K_p(x, y) = \frac{R_p(x, y, y, x)}{R_{0,p}(x, y, y, x)} = K_p.$$

Conversely, if $K(\sigma)$ is constant at point $p \in M$, that is for arbitrary x, y we have

$$R_p(x, y, y, x) = K_p R_{0,p}(x, y, y, x).$$

If we denote

$$F(s, t) = R_p(x + sz, y + tw, y + tw, x + sz) - R_p(x + sw, y + tz, y + tz, x + sw)$$

$$F_0(s, t) = R_{0,p}(x + sz, y + tw, y + tw, x + sz) - R_{0,p}(x + sw, y + tz, y + tz, x + sw),$$

we still have $F(s, t) = K_p F_0(s, t)$. By Lemma 5.15, we have

$$R_p(x, y, z, w) = -\frac{1}{6} \frac{\partial^2}{\partial s \partial t} \Big|_{s=t=0} F(s, t),$$

and it is easy to see

$$R_{0,p}(x, y, z, w) = -\frac{1}{6} \frac{\partial^2}{\partial s \partial t} \Big|_{s=t=0} F_0(s, t).$$

This completes the proof. □

Corollary 5.18. *A Riemannian manifold is isotropic if and only if $R = KR_0$, where K is a smooth function.*

Corollary 5.19. *A Riemannian manifold has constant sectional curvature K if and only if $R = KR_0$, where K is a constant.*

Remark 5.20. An important corollary is that curvature tensor of Riemannian manifold with constant sectional curvature K is quite simple, since

$$R_{ijkl} = K(g_{il}g_{jk} - g_{ik}g_{jl}),$$

that is, curvature is completely determined by zero order partial derivatives of metric, not the second order in general.

Remark 5.21. Let (M, g) be a Riemannian manifold of dimension two and $\{e_1, e_2\}$ be a basis of $T_p M$. Then

$$K_p = K_p(e_1, e_2) = \frac{R(e_1, e_2, e_2, e_1)}{|e_1|^2 |e_2|^2 - |g(e_1, e_2)|^2}$$

is called the *Gauss curvature* in the theory of surface.

5.3.2 Ricci curvature and scalar curvature

Definition 5.22. For a Riemannian manifold (M, g) , the *Ricci curvature* is defined to be

$$\text{Ric}(X, Y) := \text{tr}_g(Z \mapsto R(Z, X)Y),$$

where X, Y are vector fields.

Remark 5.23 (local form). The trace of above endomorphism is exactly R_{ijk}^i , and it can be written as

$$g^{i\ell} R_{ijk\ell}.$$

In other words, Ricci curvature tensor is the contracted tensor of curvature with respect to the first and fourth index.

Definition 5.24. For a point $p \in M$, and $x \in T_p M$, the *Ricci curvature in the direction x* is defined as

$$\text{Ric}_p(x) := \text{Ric}_p(x, x).$$

Remark 5.25. For $x \in T_p M$, we can write it as $x = x^i e_i$, where $\{e_1, \dots, e_n\}$ is a basis of $T_p M$, then

$$\text{Ric}_p(x) = R_{jk} x^j x^k.$$

Definition 5.26. For a Riemannian manifold (M, g) , the *scalar curvature S* at $p \in M$ is defined as

$$S(p) := \sum_{i=1}^n \text{Ric}_p(e_i),$$

where $\{e_1, \dots, e_n\}$ is an orthonormal basis of $T_p M$.

Remark 5.27 (local form). Locally we have

$$S = g^{jk} R_{jk}.$$

Proposition 5.28 (contracted Bianchi identity).

$$g^{jk} \nabla_k R_{ij} = \frac{1}{2} \nabla_i S,$$

where R_{ij} is Ricci curvature and S is scalar curvature.

Proof. Direct computation shows

$$\begin{aligned}
g^{jk}\nabla_k R_{ij} &= g^{jk}\nabla_k g^{pq} R_{pijq} \\
&= g^{jk} g^{pq} \nabla_k R_{pijq} \\
&= g^{jk} g^{pq} (-\nabla_p R_{ikjq} - \nabla_i R_{kpjq}) \\
&= -g^{pq} \nabla_p R_{iq} + \nabla_i S \\
&= -g^{jk} \nabla_k R_{ij} + \nabla_i S.
\end{aligned}$$

This completes the proof. $\square$

Proposition 5.29. *The scalar curvature S at $p \in M$ is given by*

$$S(p) = \frac{1}{\alpha_n} \int_{\mathbb{S}^{n-1}} \text{Ric}_p(x) d\mathbb{S}^{n-1},$$

where α_n is the volume of n -dimension unit ball in $\mathbb{R}^n$ and $d\mathbb{S}^{n-1}$ is the area elements in $\mathbb{S}^{n-1}$.

Proof. Choose an orthonormal basis $\{e_1, \dots, e_n\}$ in $T_p M$ and write $x = x^i e_i$, then

$$\text{Ric}_p(x) = \text{Ric}_p(x^i e_i) = (x^i)^2 \text{Ric}_p(e_i).$$

Since $|x| = 1$, then the vector $\mu = (x^1, \dots, x^n)$ is a unit normal vector on $\mathbb{S}^{n-1}$. Denoting $V = (x^1 \text{Ric}_p(e_1), \dots, x^n \text{Ric}_p(e_n))$, then Stokes theorem implies

$$\begin{aligned}
\frac{1}{\alpha_n} \int_{\mathbb{S}^{n-1}} (x^i)^2 \text{Ric}_p(e_i) d\mathbb{S}^{n-1} &= \frac{1}{\alpha_n} \int_{\mathbb{S}^{n-1}} \langle V, \mu \rangle d\mathbb{S}^{n-1} \\
&= \frac{1}{\alpha_n} \int_{B^n} \text{div } V dB^n \\
&= \text{div } V \\
&= \sum_{i=1}^n \text{Ric}_p(e_i) \\
&= S(p),
\end{aligned}$$

where B^n is unit ball in $T_p M$ with $\partial B^n = \mathbb{S}^{n-1}$. $\square$

Theorem 5.30. *Let (M, g) be a Riemannian manifold, then for all $p \in M$ and r sufficiently small, the volume of the geodesic ball $B(p, r)$ is*

$$\text{vol}(B(p, r)) = \alpha_n r^n \left(1 - \frac{S(p)}{6(n+2)} r^2 + O(r^3) \right),$$

where α_n is the volume of n -dimension unit ball in $\mathbb{R}^n$.

Proof. Note that we have

$$\sqrt{\det(g_{ij})} = \delta_{ij} - \frac{1}{6}R_{jk}(p)x^j x^k + O(|x|^3).$$

Directly computation shows

$$\begin{aligned} \text{Vol}(B(p, r)) &= \int_0^r \int_{\mathbb{S}^{n-1}(t)} \sqrt{\det g} dS dt \\ &= \int_0^r \int_{\mathbb{S}^{n-1}(t)} \left(1 - \frac{1}{6}\text{Ric}_p(x) + O(|x|^3)\right) dS dt \\ &= \alpha_n r^n - \frac{\alpha_n}{6} \int_0^r t^{n+1} dt + O(r^{n+3}) \\ &= \alpha_n r^n - \frac{\alpha_n S(p) r^{n+2}}{6(n+2)} + O(r^{n+3}) \\ &= \alpha_n r^n \left(1 - \frac{S(p)}{6(n+2)} r^2 + O(r^3)\right), \end{aligned}$$

where we use the fact $\alpha_n = \omega_{n-1}/n$. □

5.4 Basic models

5.4.1 Einstein manifold

Definition 5.31. A Riemannian manifold (M, g) is called *Einstein manifold*, if its Ricci curvature satisfies $R_{ij} = \lambda g_{ij}$ for some $\lambda \in \mathbb{R}$.

Lemma 5.32 (Schur's lemma). *Let (M, g) be a Riemannian manifold with $\dim M \geq 3$, suppose $R_{ij} = f g_{ij}$, where f is a smooth function, then (M, g) is an Einstein manifold.*

Proof. If $R_{ij} = f g_{ij}$, then contracted Bianchi identity shows

$$\frac{n}{2} \nabla_i f = g^{jk} \nabla_k f g_{ij} = \nabla_i f$$

for arbitrary i , which implies f is constant, since $n \geq 3$. □

Corollary 5.33. *For a Riemannian manifold (M, g) with $\dim M \geq 3$, it is isotropic if and only if it has constant sectional curvature.*

Proof. By Remark 5.14, it suffices to show if M is isotropic then it has constant sectional curvature. If M is isotropic, then there exists a smooth function K such that

$$R_{ijk\ell} = K(g_{i\ell}g_{jk} - g_{ik}g_{j\ell}).$$

Consider its Ricci curvature, that is

$$R_{jk} = (n-1)K g_{jk}.$$

Then Schur's lemma implies $(n-1)K$ is constant, that is, K is constant. □

Proposition 5.34. *Let (M, g) be an Einstein 3-manifold, then (M, g) has constant sectional curvature.*

Proof. For arbitrary point $p \in M$, without loss of generality we consider normal coordinate, that is $g_{ij} = \delta_{ij}$. Then

$$R_{11} = g^{ij} R_{i11j} = R_{2112} + R_{3113} = \lambda.$$

Similarly we have

$$R_{1221} + R_{3223} = \lambda, \quad R_{1331} + R_{2332} = \lambda.$$

Thus we can conclude

$$R_{1221} = R_{1331} = R_{2332} = \frac{\lambda}{2},$$

that is,

$$R_{ijkl} = \frac{\lambda}{2}(\delta_{il}\delta_{jk} - \delta_{ik}\delta_{jl}).$$

This shows (M, g) has constant sectional curvature $\lambda/2$. □

Remark 5.35. In fact, it is a special case of Ricci curvature controls curvature. For a n -dimensional Riemannian manifold, it is easy to see R_{jk} has $n(n+1)/2$ independent components. But for R_{ijkl} , this counting problem becomes a little bit complicated, it has

$$\frac{n^2(n^2 - 1)}{12}$$

independent components. Indeed, since R_{ijkl} is skew symmetric in ij and $k\ell$, this means that these pair of indices can take

$$m = \binom{n}{2} = \frac{n(n-1)}{2}.$$

Also, R_{ijkl} is symmetric when you swap ij with $k\ell$, which means there would be

$$\frac{m(m+1)}{2} = \frac{n^4 - 2n^3 + 3n^2 - 2n}{8}$$

choices. However, these are not independent, since there is first Bianchi identity

$$R_{ijkl} + R_{jkil} + R_{kijl} = 0,$$

and it provides

$$\binom{n}{4} = \frac{n^4 - 6n^3 + 11n^2 - 6n}{24}$$

relations between these components, thus the number of independent components of R_{ijkl} is

$$\frac{n^4 - 2n^3 + 3n^2 - 2n}{8} - \frac{n^4 - 6n^3 + 11n^2 - 6n}{24} = \frac{n^4 - n^2}{12} = \frac{n^2(n^2 - 1)}{12}.$$

Therefore curvature is fully determined by the Ricci curvature if and only if

$$\frac{n^2(n^2 - 1)}{12} \leq \frac{n(n+1)}{2},$$

or in other words, $n \leq 3$.

5.4.2 Lie group with invariant metric

Lemma 5.36. *Let G be a Lie group equipped with left-invariant metric $\langle \cdot, \cdot \rangle$, and ∇ the Levi-Civita connection with respect to it. Then for all left-invariant vector fields X, Y, Z ,*

$$\langle X, \nabla_Y Z \rangle = \frac{1}{2} (\langle [X, Y], Z \rangle + \langle [X, Z], Y \rangle + \langle [Y, Z], X \rangle).$$

Proof. Recall that

$$\langle X, \nabla_Y Z \rangle = \frac{1}{2} (Y \langle Z, X \rangle + Z \langle X, Y \rangle - X \langle Y, Z \rangle - \langle [Y, X], Z \rangle - \langle [Z, X], Y \rangle - \langle [Y, Z], X \rangle).$$

But $Y \langle Z, X \rangle = Z \langle X, Y \rangle = X \langle Y, Z \rangle = 0$ since both metric and X, Y, Z are left-invariant, that is,

$$\langle X, \nabla_Y Z \rangle = \frac{1}{2} (\langle [X, Y], Z \rangle + \langle [X, Z], Y \rangle + \langle [Y, Z], X \rangle).$$

Now set $Y = Z$ to conclude. □

Proposition 5.37. *Let G be a Lie group equipped with bi-invariant metric $\langle \cdot, \cdot \rangle$, and ∇ the Levi-Civita connection with respect to it. Then for all left-invariant vector fields X, Y, Z ,*

$$\langle [X, Y], Z \rangle = \langle X, [Y, Z] \rangle.$$

Proof. Let y_t be the flow of Y , then

$$[X, Y] = \lim_{t \rightarrow 0} \frac{1}{t} ((dy_t)(X) - X).$$

On the other hand, since Y is left-invariant, that is $L_g \circ y_t = y_t \circ L_g$, giving

$$y_t(g) = y_t(L_g(e)) = L_g y_t(e) = g y_t(e) = R_{y_t(e)}(g).$$

Thus $dy_t = dR_{y_t(e)}$ and

$$[X, Y] = \lim_{t \rightarrow 0} \frac{1}{t} ((dR_{y_t(e)})(X) - X).$$

Note that the metric is bi-invariant, thus

$$\langle X, Z \rangle = \langle (dR_{y_t(e)}) \circ (dL_{y_t^{-1}(e)}) X, (dR_{y_t(e)}) \circ (dL_{y_t^{-1}(e)}) Z \rangle = \langle (dR_{y_t(e)}) X, (dR_{y_t(e)}) Z \rangle.$$

Differentiating the expression above with respect to t and setting $t = 0$ we conclude

$$0 = \langle [X, Y], Z \rangle + \langle X, [Z, Y] \rangle.$$

This completes the proof. □

Theorem 5.38. *Let G be a Lie group equipped with bi-invariant metric $\langle \cdot, \cdot \rangle$, and ∇ the Levi-Civita connection with respect to it. Then for every left-invariant vector field X on G , then $\nabla_X X = 0$.*

Proof. From Lemma 5.36, we have

$$\langle Y, \nabla_X X \rangle = \langle X, [Y, X] \rangle,$$

From Proposition 5.37, we have

$$\langle X, [Y, X] \rangle = \langle [X, Y], X \rangle = -\langle X, [Y, X] \rangle,$$

that is, $\langle Y, \nabla_X X \rangle = 0$ for arbitrary vector field Y , which implies $\nabla_X X = 0$. $\square$

Corollary 5.39. *The assumptions are as above. If X, Y are left-invariant vector fields, then $\nabla_X Y = \frac{1}{2}[X, Y]$.*

Proof. Note that

$$\begin{aligned} 0 &= \nabla_{X+Y}(X+Y) \\ &= \nabla_X Y + \nabla_Y X + \nabla_X X + \nabla_Y Y \\ &= \nabla_X Y + \nabla_Y X \\ &= 2\nabla_X Y - [X, Y]. \end{aligned}$$

Division by two finally yields

$$\nabla_X Y = \frac{1}{2}[X, Y].$$

$\square$

Corollary 5.40. *The assumptions are as above. If X, Y, Z are left-invariant vector fields, then $R(X, Y)Z = -\frac{1}{4}[[X, Y], Z]$.*

Proof. Directly from $\nabla_X Y = \frac{1}{2}[X, Y]$ and Jacobi's identity. $\square$

Corollary 5.41. *The assumptions are as above. If X, Y are left-invariant vector fields which are orthogonal, and σ is the plane generated by X and Y . Then*

$$K(\sigma) = \frac{1}{4}||[X, Y]||^2.$$

Proof.

$$\begin{aligned} K(\sigma) &= \frac{\langle R(X, Y)Y, X \rangle}{||X||^2 ||Y||^2 - \langle X, Y \rangle^2} \\ &= \frac{\langle -\frac{1}{4}[[X, Y], Y], X \rangle}{||X||^2 ||Y||^2} \\ &= -\frac{1}{4} \langle [[X, Y], Y], X \rangle \\ &= -\frac{1}{4} \langle [X, Y], [Y, X] \rangle \\ &= \frac{1}{4} ||[X, Y]||^2. \end{aligned}$$

$\square$

Remark 5.42. Therefore, sectional curvature of a Lie group with bi-invariant metric is non-negative. Furthermore, if the center of Lie algebra $\mathfrak{g}$ is trivial, then the sectional curvature is positive.

Theorem 5.43. *Let G be a Lie group equipped with bi-invariant metric, the geodesics on G are precisely the integral curves of left-invariant vector fields.*

Proof. Let $X \in \mathfrak{g}$ be a left-invariant vector field, and $\gamma : \mathbb{R} \rightarrow G$ its integral curve. Then

$$\nabla_{\frac{d}{dt}} \gamma_* \left(\frac{d}{dt} \right) = \nabla_{\gamma_* \left(\frac{d}{dt} \right)} \gamma_* \left(\frac{d}{dt} \right) = \nabla_X X = 0,$$

which implies integral curves of left-invariant vector fields are geodesics. Furthermore, since geodesics are unique, we have geodesics are precisely integral curves of left-invariant vector fields. $\square$

Corollary 5.44. *The exponential map for the Lie group coincides with the exponential map of the Levi-Civita connection with respect to bi-invariant metric.*

6 Hodge theory on Riemannian manifold

In this section, we always assume (M, g) is a compact oriented Riemannian n -manifold, as we need to consider integrations on M .

6.1 Inner product on differential forms

Before we introduce the Hodge theory on (M, g) , let's recall some basic facts about differential k -forms. For a k -form φ , locally it can be written as

$$\varphi = \sum_{1 \leq i_1 < \dots < i_k \leq n} \varphi_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k},$$

where $\varphi_{i_1 \dots i_k} := \varphi\left(\frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_k}}\right)$ is a skew-symmetric function. Equivalently, we have

$$\varphi = \frac{1}{k!} \varphi_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

Notation 6.1. For convenience, we use $\varphi_I dx^I$ to denote the summation $\varphi_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}$.

Definition 6.2. Let φ, ψ be two k -forms. The *metric pairing* $\langle -, - \rangle$ on Ω_M^k is defined as

$$\langle \varphi, \psi \rangle := \frac{1}{k!} g(\varphi, \psi),$$

where g is induced metric on $\Omega_M^k \subseteq \bigotimes^k T^*M$.

Lemma 6.3. For $\varphi = \varphi_I dx^I, \psi = \psi_J dx^J$, we have

$$\langle \varphi, \psi \rangle = \varphi_I \psi_J g^{IJ},$$

where

$$g^{IJ} = \frac{1}{k!} g(dx^I, dx^J) = \det \begin{pmatrix} g^{i_1 j_1} & \dots & g^{i_1 j_k} \\ \dots & \dots & \dots \\ g^{i_k j_1} & \dots & g^{i_k j_k} \end{pmatrix}.$$

Proof. It suffices to show

$$g(dx^I, dx^J) = k! \det \begin{pmatrix} g^{i_1 j_1} & \dots & g^{i_1 j_k} \\ \dots & \dots & \dots \\ g^{i_k j_1} & \dots & g^{i_k j_k} \end{pmatrix}.$$

By definition one has

$$dx^I = dx^{i_1} \wedge \dots \wedge dx^{i_k} = \sum_{\sigma \in S_k} (-1)^{|\sigma|} dx^{i_{\sigma(1)}} \otimes \dots \otimes dx^{i_{\sigma(k)}}.$$

Then

$$\begin{aligned} g(dx^I, dx^J) &= \sum_{\sigma, \tau} (-1)^{|\sigma|} (-1)^{|\tau|} g(dx^{i_{\sigma(1)}} \otimes \dots \otimes dx^{i_{\sigma(k)}}, dx^{j_{\tau(1)}} \otimes \dots \otimes dx^{j_{\tau(k)}}) \\ &= \sum_{\sigma, \tau} (-1)^{|\sigma|} (-1)^{|\tau|} g^{i_{\sigma(1)} j_{\tau(1)}} \dots g^{i_{\sigma(k)} j_{\tau(k)}} \\ &= \sum_{\sigma, \tau} (-1)^{|\sigma \tau^{-1}|} g^{i_{\sigma \tau^{-1}(1)} j_1} \dots g^{i_{\sigma \tau^{-1}(k)} j_k} \\ &= \sum_{\sigma} \sum_{\rho} (-1)^{|\rho|} g^{i_{\rho(1)} j_1} \dots g^{i_{\rho(k)} j_k} \\ &= \sum_{\sigma} \det(g^{i_p j_q}) \\ &= k! \det(g^{i_p j_q}). \end{aligned}$$

□

Corollary 6.4. For two k -forms φ, ψ , locally written as

$$\begin{aligned} \varphi &= \sum_{1 \leq i_1 < \dots < i_k \leq n} \varphi_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} \\ \psi &= \sum_{1 \leq j_1 < \dots < j_k \leq n} \psi_{j_1 \dots j_k} dx^{j_1} \wedge \dots \wedge dx^{j_k} \end{aligned}$$

with φ_I, ψ_J is skew-symmetric. Then

$$\langle \varphi, \psi \rangle = \sum_{\substack{1 \leq i_1 < \dots < i_k \leq n \\ 1 \leq j_1 < \dots < j_k \leq n}} \varphi_{i_1 \dots i_k} \psi_{j_1 \dots j_k} \det \begin{pmatrix} g^{i_1 j_1} & \dots & g^{i_1 j_k} \\ \dots & \dots & \dots \\ g^{i_k j_1} & \dots & g^{i_k j_k} \end{pmatrix}.$$

Example 6.5. Let φ, ψ be two 2-forms, locally written as

$$\begin{aligned}\varphi &= \varphi_{i_1 i_2} dx^{i_1} \wedge dx^{i_2} \\ \psi &= \psi_{j_1 j_2} dx^{j_1} \wedge dx^{j_2},\end{aligned}$$

where $i_1 < i_2, j_1 < j_2$. Then

$$\begin{aligned}\langle \varphi, \psi \rangle &= \frac{1}{2} \varphi_{i_1 i_2} \psi_{j_1 j_2} g(dx^{i_1} \wedge dx^{i_2}, dx^{j_1} \wedge dx^{j_2}) \\ &= \frac{1}{2} \varphi_{i_1 i_2} \psi_{j_1 j_2} g(dx^{i_1} \otimes dx^{i_2} - dx^{i_2} \otimes dx^{i_1}, dx^{j_1} \otimes dx^{j_2} - dx^{j_2} \otimes dx^{j_1}) \\ &= \frac{1}{2} \varphi_{i_1 i_2} \psi_{j_1 j_2} (g^{i_1 j_1} g^{i_2 j_2} - g^{i_1 j_2} g^{i_2 j_1} - g^{i_2 j_1} g^{i_1 j_2} + g^{i_2 j_2} g^{i_1 j_1}) \\ &= \varphi_{i_1 i_2} \psi_{j_1 j_2} (g^{i_1 j_1} g^{i_2 j_2} - g^{i_1 j_2} g^{i_2 j_1}) \\ &= \varphi_{i_1 i_2} \psi_{j_1 j_2} \det \begin{pmatrix} g^{i_1 j_1} & g^{i_1 j_2} \\ g^{i_2 j_1} & g^{i_2 j_2} \end{pmatrix}.\end{aligned}$$

Definition 6.6. The *volume form* vol is an n -form such that $\langle \text{vol}, \text{vol} \rangle = 1$.

Remark 6.7 (local form). The volume form is locally given by $\sqrt{\det g} dx^1 \wedge \dots \wedge dx^n$.

Definition 6.8. For two k -forms φ, ψ , the *inner product* is defined as

$$(\varphi, \psi) := \int_M \langle \varphi, \psi \rangle \text{vol}.$$

Definition 6.9. For a k -form φ and a $(k+1)$ -form ψ , the *formal adjoint* of d is an operator $d^* : C^\infty(M, \Omega_M^{k+1}) \rightarrow C^\infty(M, \Omega_M^k)$ such that

$$(d\varphi, \psi) = (\varphi, d^*\psi).$$

Remark 6.10. There is no guarantee for existence, but later we will see such d^* do exist, and give an explicit formula.

Definition 6.11. The *Laplace-Beltrami operator* $\Delta_g : C^\infty(M, \Omega_M^k) \rightarrow C^\infty(M, \Omega_M^k)$ is defined as

$$\Delta_g = dd^* + d^*d.$$

Definition 6.12. A k -form α is called *harmonic*, if $\Delta_g \alpha = 0$.

Notation 6.13. The space of all harmonic forms is denoted by $\mathcal{H}^k(M)$.

Lemma 6.14. A k -form α is harmonic if and only if $d\alpha = 0$ and $d^*\alpha = 0$.

Proof. Note that

$$\begin{aligned}(\alpha, \Delta\alpha) &= (\alpha, dd^*\alpha) + (\alpha, d^*d\alpha) \\ &= \|d^*\alpha\|^2 + \|d\alpha\|^2.\end{aligned}$$

□

6.2 Hodge star operator

Although we have defined an inner product on Ω_M^k , it is still quite complicated to compute. However, inner product on Ω_M^k is independent of the choice of local frame, so we can use normal coordinate to give a local frame, and define the Hodge star operator on it, which will give us an effective method to compute.

6.2.1 Baby case

Let V be an $\mathbb{R}$ -vector space with inner product $\langle -, - \rangle$, and $\{e_1, \dots, e_n\}$ is a basis of V . For any $0 \leq k \leq n$, there is a natural basis of $\bigwedge^k V$, consisting of $\{e_I := e_{i_1} \wedge \dots \wedge e_{i_k} \mid 1 \leq i_1 < \dots < i_k \leq n\}$.

With respect to this basis, the induced metric on $\bigwedge^k V$ can be written as

$$\langle e_{i_1} \wedge \dots \wedge e_{i_k}, e_{j_1} \wedge \dots \wedge e_{j_k} \rangle = \det \begin{pmatrix} \langle e_{i_1}, e_{j_1} \rangle & \dots & \langle e_{i_1}, e_{j_k} \rangle \\ \vdots & & \vdots \\ \langle e_{i_k}, e_{j_1} \rangle & \dots & \langle e_{i_k}, e_{j_k} \rangle \end{pmatrix}.$$

It is clear if $\{e_1, \dots, e_n\}$ is an orthonormal basis of V , then $\{e_I\}$ is an orthonormal basis of $\bigwedge^k V$. From now on, we assume $\{e_I\}$ is an orthonormal basis of $\bigwedge^k V$ with respect to inner product induced from the inner product on V .

Definition 6.15 (Hodge star). The Hodge star operator is a $\mathbb{R}$ -linear operator defined as

$$\begin{aligned} \star: \bigwedge^k V &\rightarrow \bigwedge^{n-k} V \\ e_I &\mapsto \text{sign}(I, I^c) e_{I^c}, \end{aligned}$$

where I^c is $\{1, 2, \dots, n\} \setminus I = \{i'_1, \dots, i'_{n-k}\}$ and $\text{sign}(I, I^c)$ is the sign of the permutation $(i_1, \dots, i_k, i'_1, \dots, i'_{n-k})$.

Example 6.16. It is clear $\star 1 = \text{vol}$ and $\star \text{vol} = 1$.

Proposition 6.17.

$$\star^2 = (-1)^{k(n-k)} \text{id}$$

holds on $\bigwedge^k V$.

Proof. It suffices to check on basis e_I as follows

$$\begin{aligned} \star^2 e_I &= \star(\text{sign}(I, I^c) e_{I^c}) \\ &= \text{sign}(I, I^c) \text{sign}(I^c, I) e_I \\ &= (-1)^{k(n-k)} e_I, \end{aligned}$$

where $\text{sign}(I, I^c) \text{sign}(I^c, I) = (-1)^{k(n-k)}$ is an elementary combinatorial identity. $\square$

Proposition 6.18. For $u \in \bigwedge^k V, v \in \bigwedge^{n-k} V$, we have

$$\star(u \wedge v) = (-1)^{k(n-k)} \langle u, \star v \rangle.$$

Proof. It suffices to check on basis. Without lose of generality, we may assume $u = e_I = e_{i_1} \wedge \cdots \wedge e_{i_k}$ and $v = e_{I^c}$. A direct computation shows

$$\begin{aligned} \star(e_I \wedge e_{I^c}) &= \star(\text{sign}(I, I^c) \text{vol}) \\ &= \text{sign}(I, I^c). \\ \langle e_I, \star e_{I^c} \rangle &= \langle e_I, \text{sign}(I, I^c) e_I \rangle \\ &= \text{sign}(I, I^c) \langle e_I, e_I \rangle \\ &= \text{sign}(I, I^c). \end{aligned}$$

This completes the proof. □

Corollary 6.19. For $u, v \in \bigwedge^k V$, we have

- (1) $u \wedge \star v = v \wedge \star u = \langle u, v \rangle \text{vol}.$
- (2) $\langle \star u, \star v \rangle = \langle u, v \rangle$, that is, Hodge star is a linear isometry.

Proof. For (1).

$$\star(u \wedge \star v) = (-1)^{k(n-k)} \langle u, \star^2 v \rangle = \langle u, v \rangle,$$

which implies $u \wedge \star v = \langle u, v \rangle \text{vol}$. Since $\langle u, v \rangle = \langle v, u \rangle$, we obtain $u \wedge \star v = v \wedge \star u$.

For (2).

$$\begin{aligned} \langle \star u, \star v \rangle &= (-1)^{k(n-k)} \star(\star u \wedge v) \\ &= (-1)^{2k(n-k)} \star(v \wedge \star u) \\ &= (-1)^{3k(n-k)} \langle v \wedge \star^2 u \rangle \\ &= (-1)^{4k(n-k)} \langle v, u \rangle \\ &= \langle u, v \rangle. \end{aligned}$$

□

Corollary 6.20. For $u \in \bigwedge^k V, v \in \bigwedge^{n-k} V$, we have

$$\langle u, \star v \rangle = (-1)^{k(n-k)} \langle \star u, v \rangle.$$

Proof.

$$\langle u, \star v \rangle = \langle \star u, \star^2 v \rangle = (-1)^{k(n-k)} \langle \star u, v \rangle.$$

□

Remark 6.21. This corollary implies the adjoint operator of $\star$ is $(-1)^{k(n-k)} \star$.

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